Managerial Allocation Skill and the Value of Digital Monitoring

Yuyu Chen[†] Miguel Espinosa[‡] Shuo Liu[†] Jingchen Zhu[†]

[†]Peking University

[‡]Bocconi University, CEPR, LEAP and CESifo

Fall 2026

Digital monitoring technologies



Benefits and problems

- **Benefits:** ↑ managerial span of control, ↑ accountability, ↑ procedural compliance, ↓ moral hazard
  - (Bloom et al., 2014; Pierce et al., 2015; Staats et al., 2017; Kelley et al., 2024; Ko and Baek, 2025)

- **Problems:** crowd out intrinsic motivation, weaken cooperation, and encourage narrow responses to measured targets
  - (Frey, 1993; Falk and Kosfeld, 2006; Blader et al., 2020; Ranganathan and Benson, 2020)

Research question

When does digital monitoring create value, and how does managerial skill shape that value?

Can DM be used well?

- Digital monitoring gives managers richer information about workers and tasks.
  - real-time task progress
  - worker-specific performance
  - task completion and assignment records

Can DM be used well?

- Digital monitoring gives managers richer information about workers and tasks.
  - real-time task progress
  - worker-specific performance
  - task completion and assignment records
- But how that information is used?
  - managers use it to *improve orchestration*
  - but not all managers really know *how to improve*

Can DM be used well?

- Digital monitoring gives managers richer information about workers and tasks.
  - real-time task progress
  - worker-specific performance
  - task completion and assignment records
- But how that information is used?
  - managers use it to *improve orchestration*
  - but not all managers really know *how to improve*
- **The value of monitoring may therefore depend on managerial skill.**

This paper

- Partnership with a large manufacturing firm
  - Organized into 38 production teams
  - about 800 production workers
  - Since 2020, it has been rolling out a **digital tracking system**.

- The digital tracking system does not increase average worker productivity
  - total output ↑ as managers oversee larger teams

- **Main finding:** productivity of teams with good managers ↑, while those with other managers ↓
  - Main mechanism: better good worker retention
  - GW with GM: Mistake ↓, absense ↓, feel overburdening ↓

This paper

- Partnership with a large manufacturing firm
  - Organized into 38 production teams
  - about 800 production workers
  - Since 2020, it has been rolling out a **digital tracking system**.

- The digital tracking system does not increase average worker productivity
  - total output ↑ as managers oversee larger teams

- **Main finding:** productivity of teams with good managers ↑, while those with other managers ↓
  - Main mechanism: better good worker retention
  - GW with GM: Mistake ↓, absense ↓, feel overburdening ↓

This paper

- Partnership with a large manufacturing firm
  - Organized into 38 production teams
  - about 800 production workers
  - Since 2020, it has been rolling out a **digital tracking system**.

- The digital tracking system does not increase average worker productivity
  - total output ↑ as managers oversee larger teams

- **Main finding:** productivity of teams with good managers ↑, while those with other managers ↓
  - Main mechanism: better good worker retention
  - GW with GM: Mistake ↓, absense ↓, feel overburdening ↓

The tracking system

- How it works?
  - Team managers make production plans and assign tasks to workers;
  - Print the QR codes for each task;
  - Workers scan the codes to report task start;
  - Workers do the tasks;
  - Workers scan the codes to report task finish.
- Direct impact
  - allow team managers to access real time production information
  - clearer evaluation of workers
- Staggered implementation
  - By the end of 2023, 35 out of 38 teams had implemented the system
  - Team managers have no power in deciding when to implement

Staggered Implementation Balance Tests

The tracking system

- How it works?
  - Team managers make production plans and assign tasks to workers;
  - Print the QR codes for each task;
  - Workers scan the codes to report task start;
  - Workers do the tasks;
  - Workers scan the codes to report task finish.
- Direct impact
  - allow team managers to access real time production information
  - clearer evaluation of workers
- Staggered implementation
  - By the end of 2023, 35 out of 38 teams had implemented the system
  - Team managers have no power in deciding when to implement

Staggered Implementation Balance Tests

The tracking system

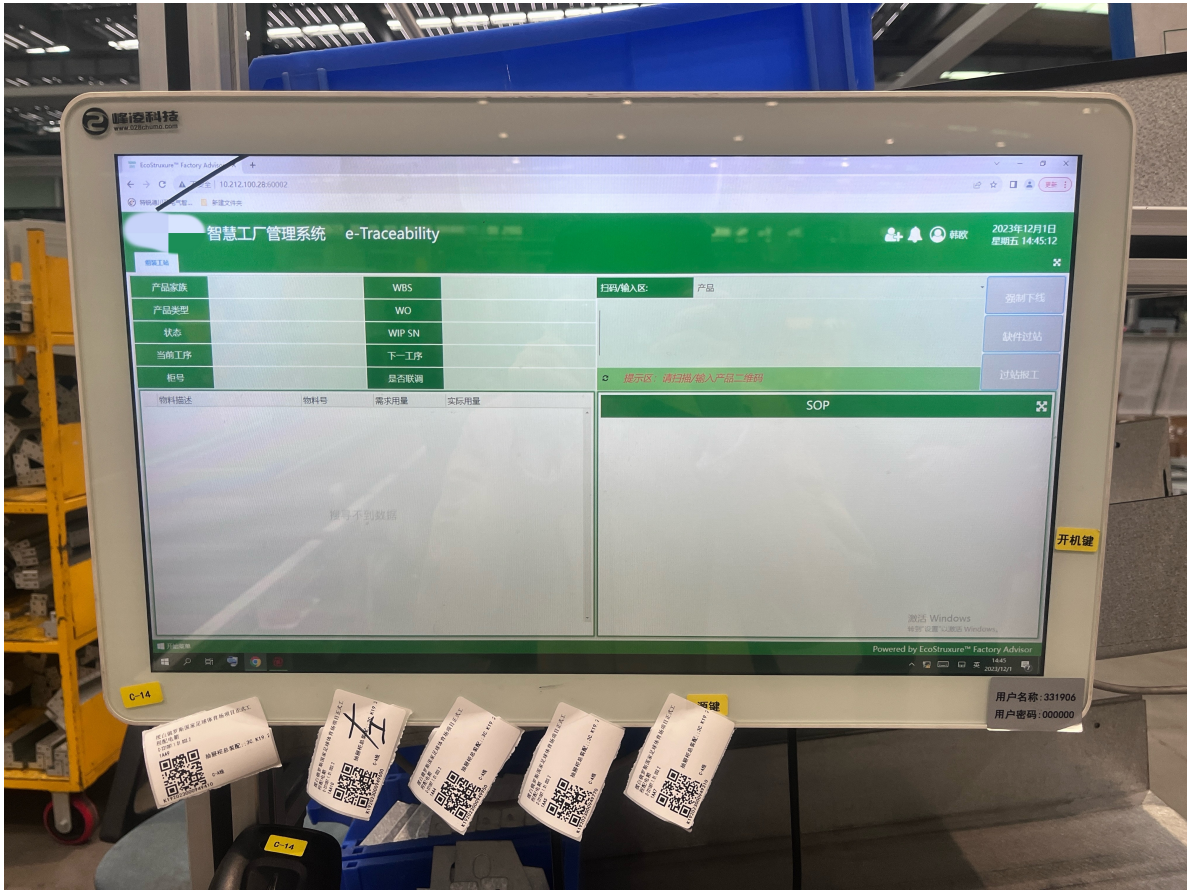
- How it works?
  - Team managers make production plans and assign tasks to workers;
  - Print the QR codes for each task;
  - Workers scan the codes to report task start;
  - Workers do the tasks;
  - Workers scan the codes to report task finish.
- Direct impact
  - allow team managers to access real time production information
  - clearer evaluation of workers
- Staggered implementation
  - By the end of 2023, 35 out of 38 teams had implemented the system
  - Team managers have no power in deciding when to implement

▸ Staggered Implementation ▸ Balance Tests

Manager using the system



Workers' QR codes for tasks



Classifying workers and managers

- **Good workers:** workers whose performance wage often ranks among the top 50% within their teams Details
- **Good managers:** managers whose average leader allowance ranks within top 25% Details
  - leader allowance: default 300 CNY, adjustable
  - reflect senior managements' evaluation for team managers' managerial performance
- **Teams with good managers:** teams having at least one good manager before implementation Details
  - 16 out of 38

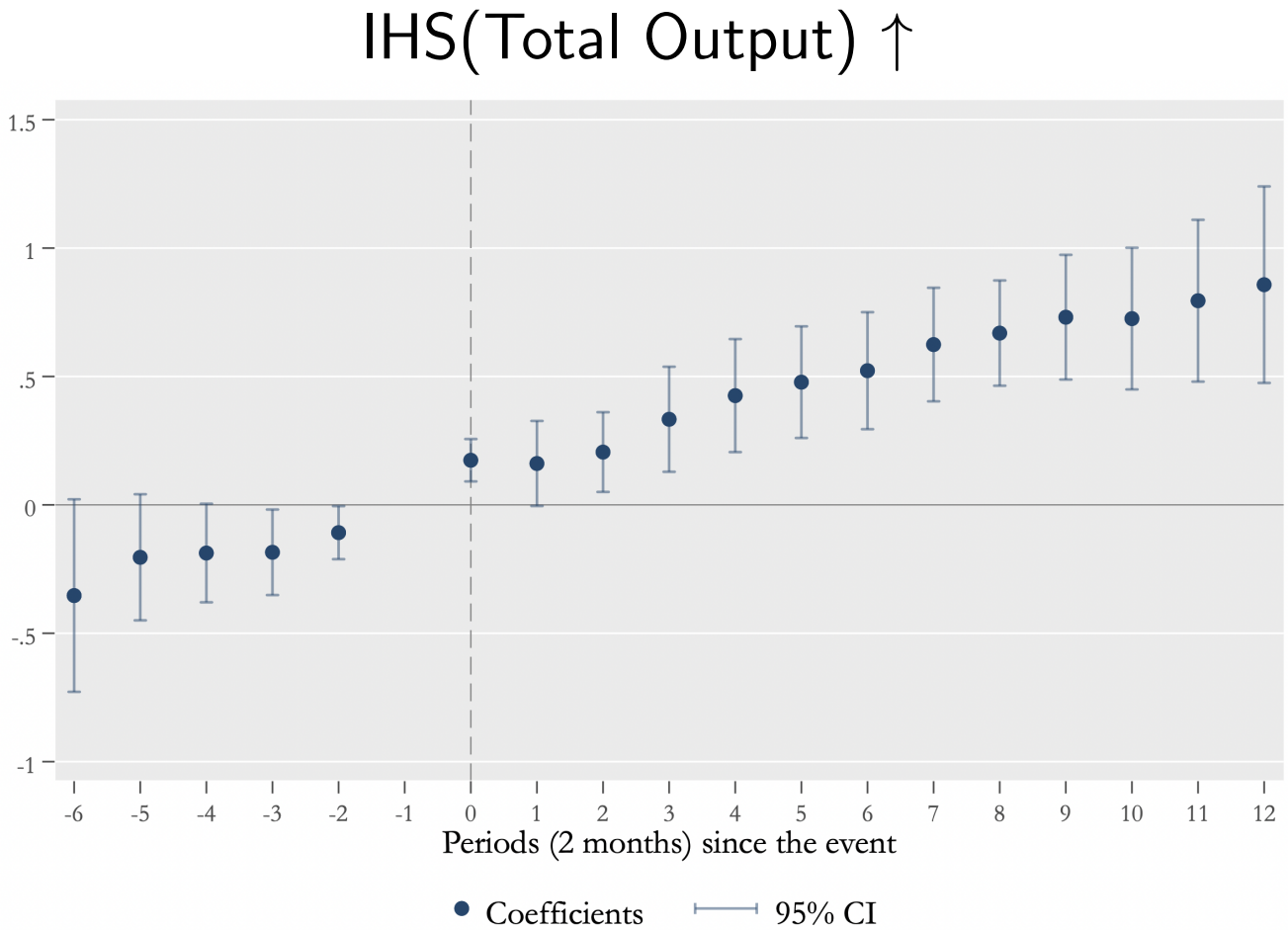
Classifying workers and managers

- **Good workers:** workers whose performance wage often ranks among the top 50% within their teams Details
- **Good managers:** managers whose average leader allowance ranks within top 25% Details
  - leader allowance: default 300 CNY, adjustable
  - reflect senior managements' evaluation for team managers' managerial performance
- **Teams with good managers:** teams having at least one good manager before implementation Details
  - 16 out of 38

Classifying workers and managers

- **Good workers:** workers whose performance wage often ranks among the top 50% within their teams Details
- **Good managers:** managers whose average leader allowance ranks within top 25% Details
  - leader allowance: default 300 CNY, adjustable
  - reflect senior managements' evaluation for team managers' managerial performance
- **Teams with good managers:** teams having at least one good manager before implementation Details
  - 16 out of 38

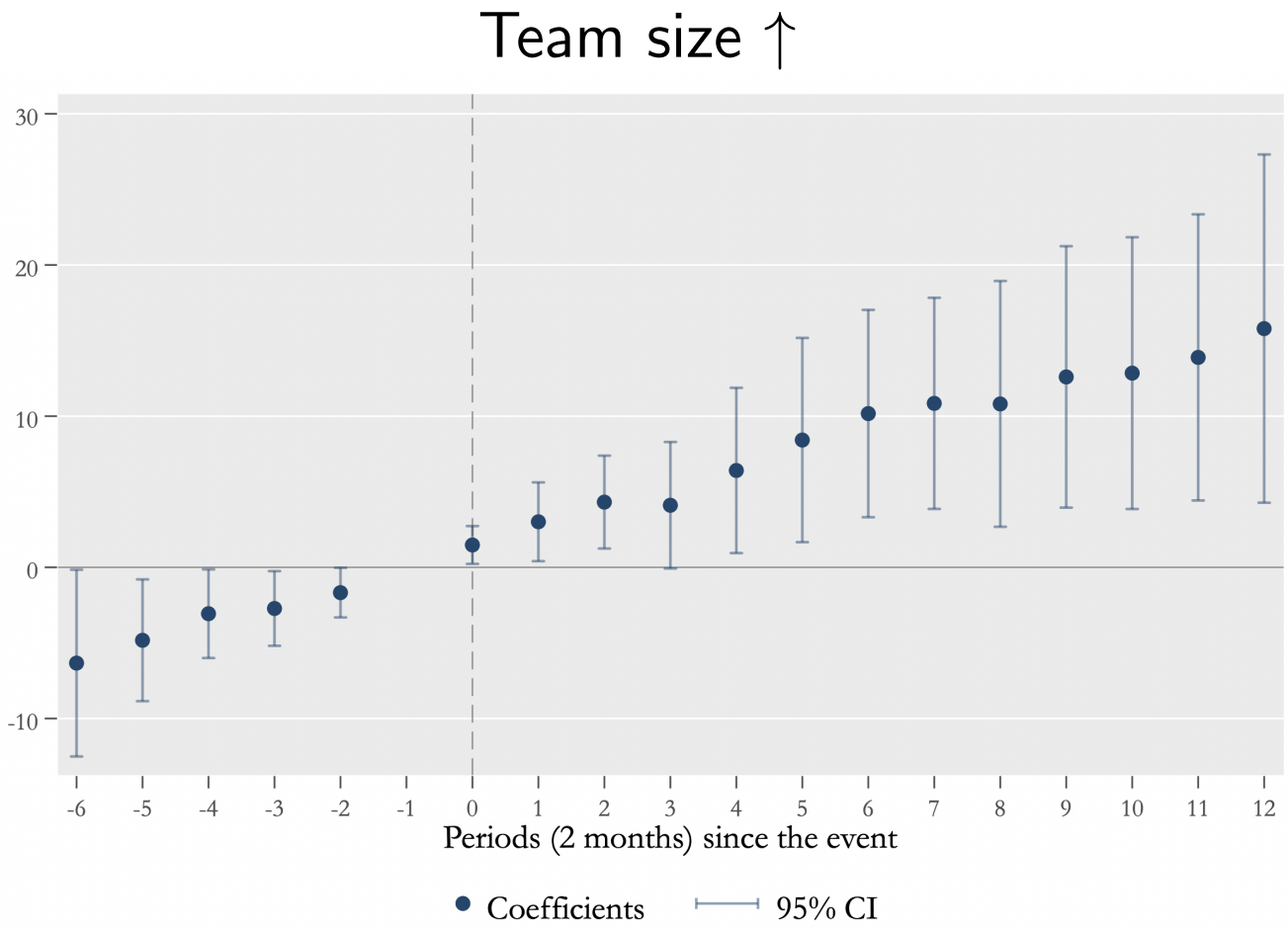
Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Specification ▸ Other DVs

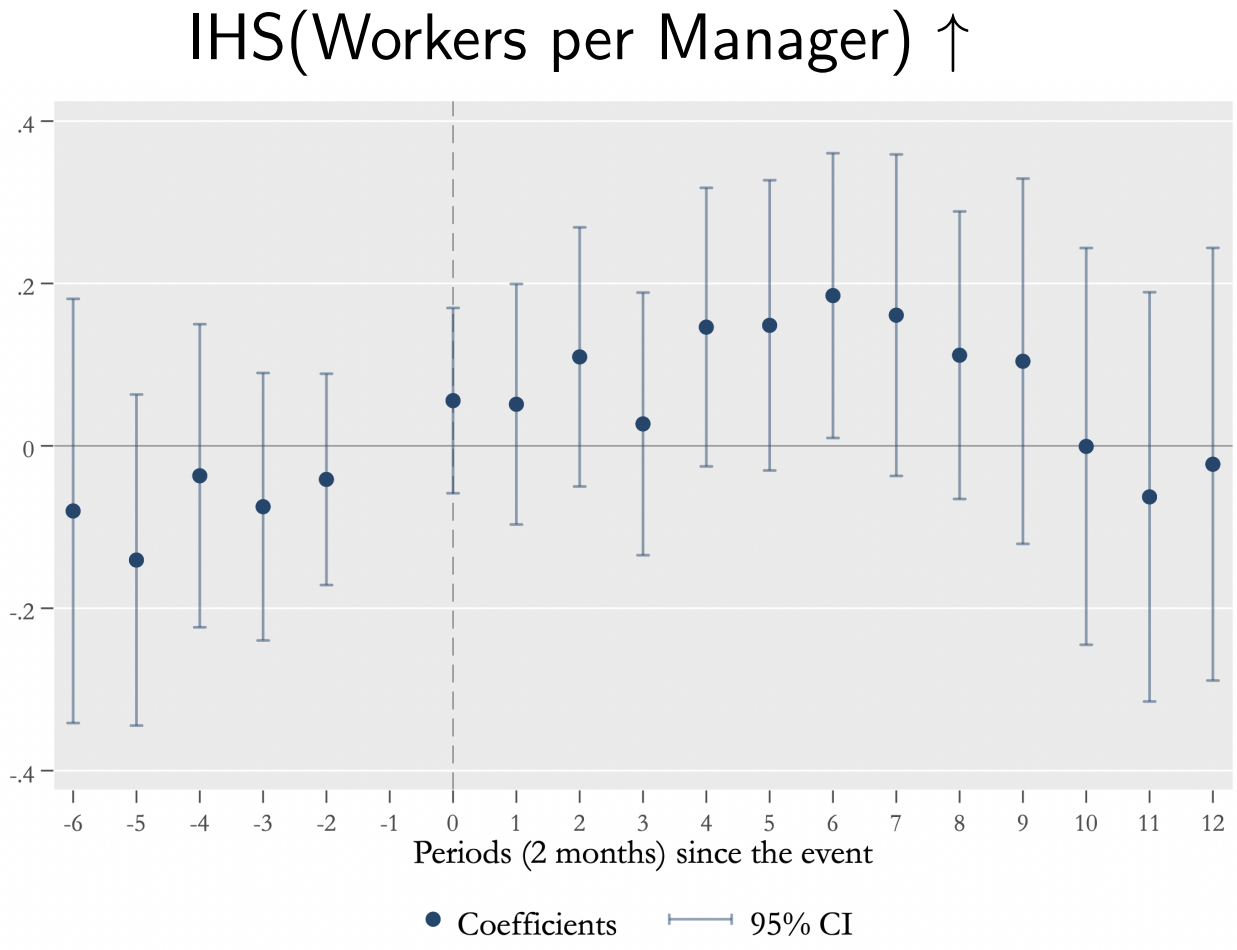
Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Specification ▸ Other DVs

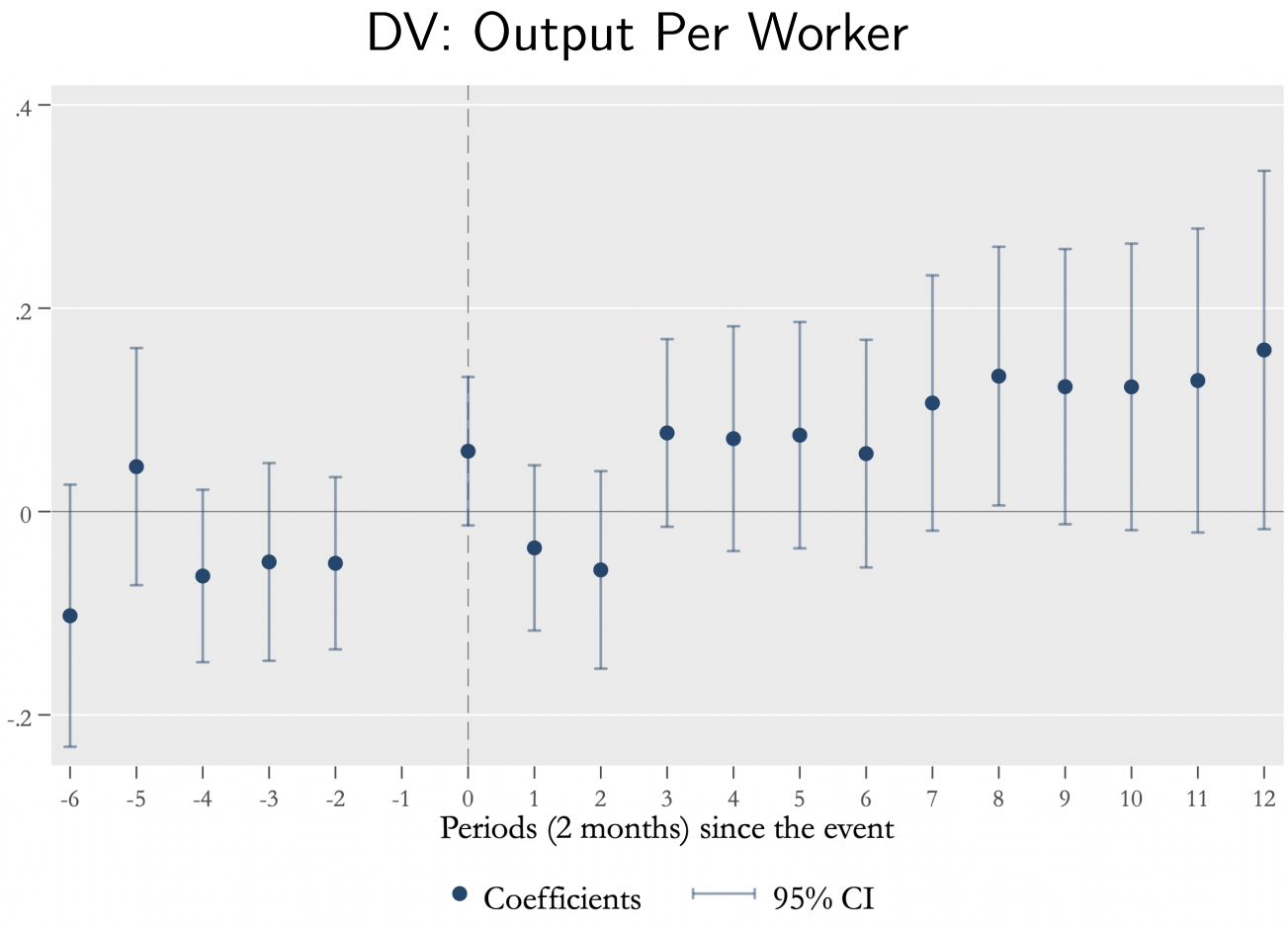
Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Specification ▸ Other DVs

No significant impact on productivity?

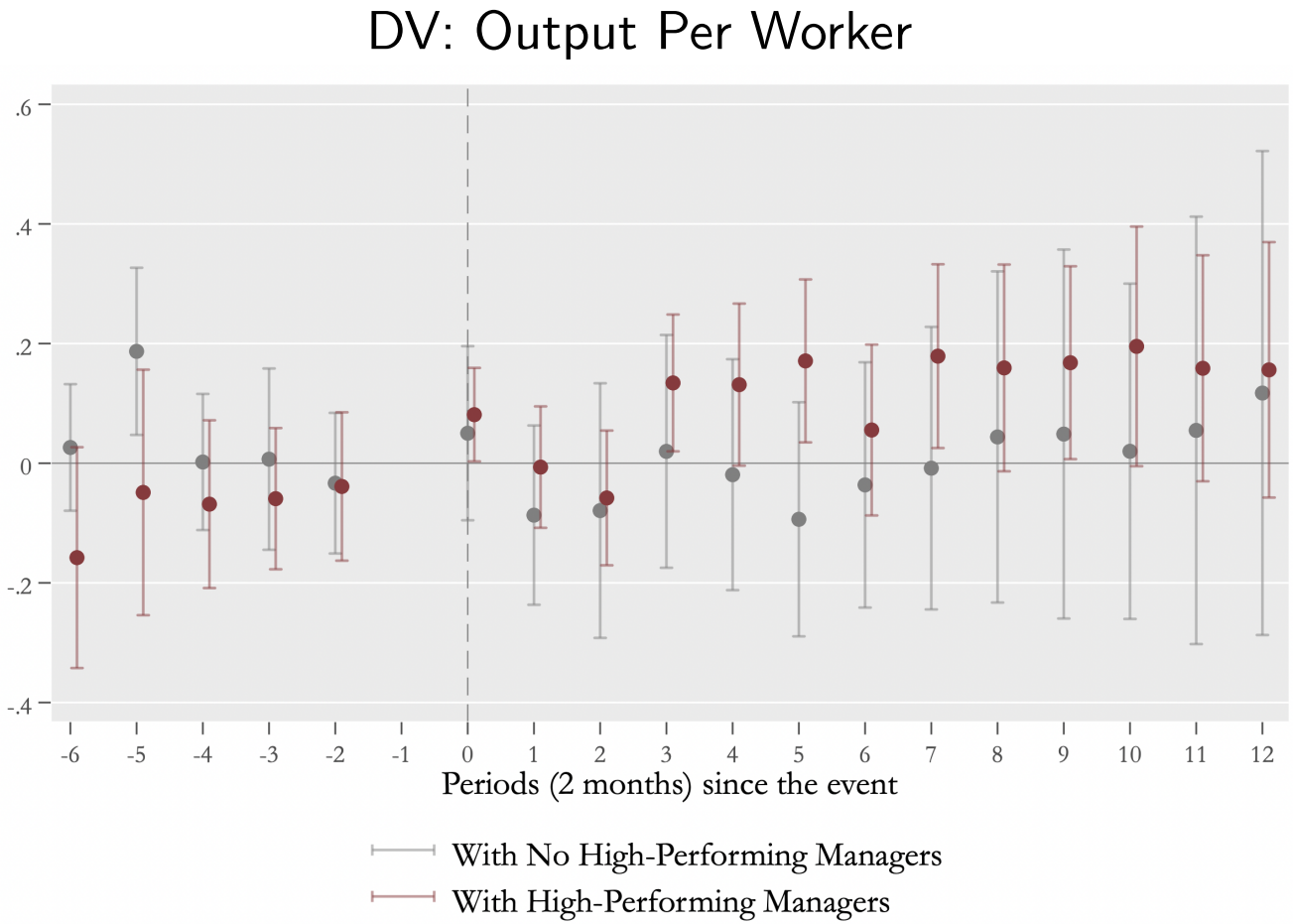


Controls for: demographics + team FE + period FE and SE clustered at team level

▸ Specification ▸ Other DVs

Heterogeneous Effects by Managerial Quality

Productivity gains only for good managers

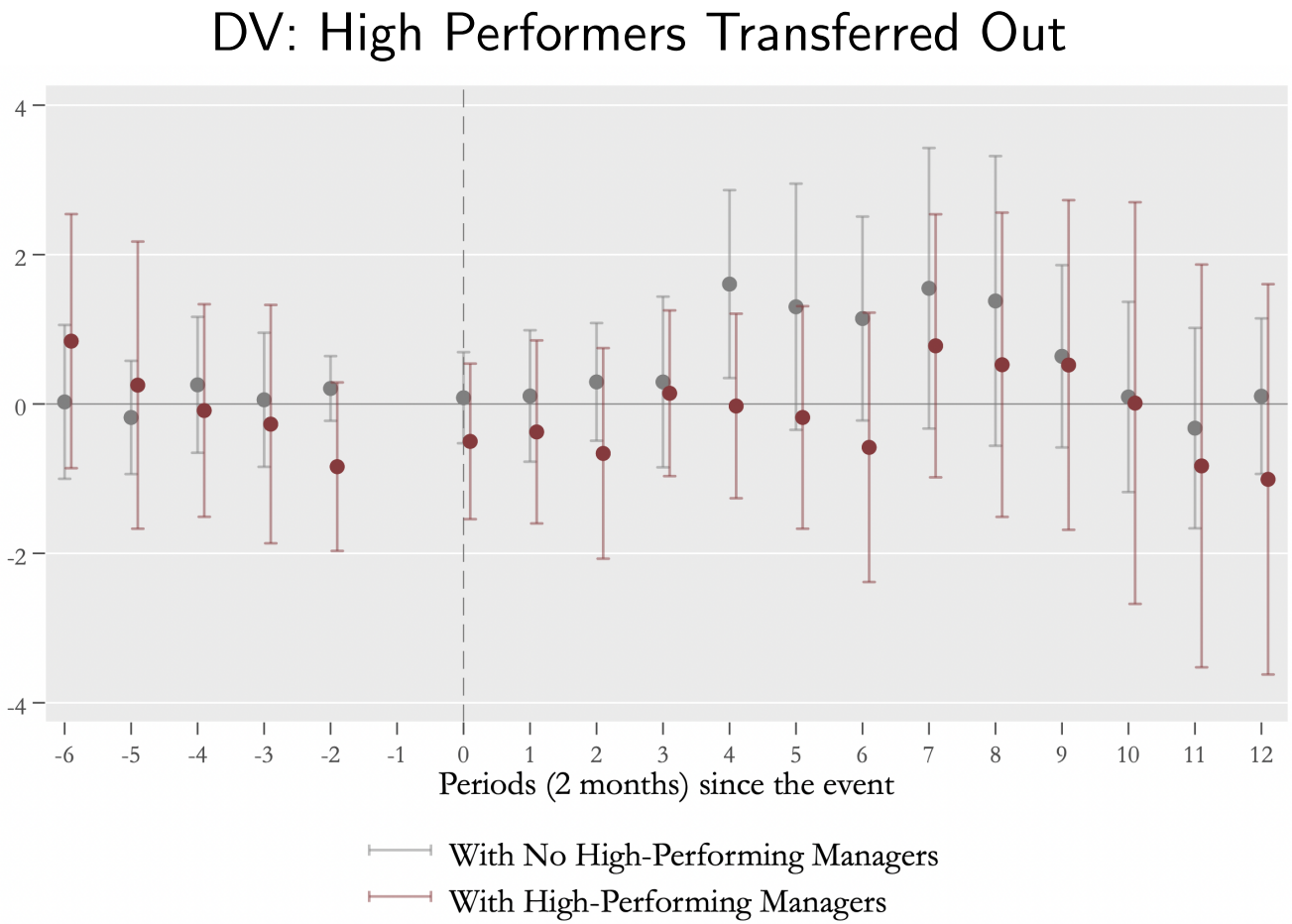


- Teams without GM: -0.063 (SE 0.030), DiD coefficient: 0.179 (SE 0.066)

Controls for: demographics + team FE + period FE and SE clustered at team level.

▸ Specification ▸ Other DVs

Higher retention for good managers



Controls for: demographics + team FE + period FE and SE clustered at team level

▸ Specification ▸ Other DVs

Higher retention for good managers

Table: Heterogeneous Effects by Managerial Quality

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)	(3) High Performers Transferred Out
Implemented	0.367***	-0.063**	1.040*
	(0.121)	(0.030)	(0.632)
Implemented × High-Performing Manager	-0.018	0.179**	-1.713**
	(0.212)	(0.066)	(0.816)
Observations	955	955	642
R-squared	0.819	0.697	
Control Mean	12.28	9.83	0.42
Team Controls	Yes	Yes	Yes
Team FE	Yes	Yes	Yes
Group*Period FE	Yes	Yes	Yes

Notes. Obs. at team-bimontly period level. Column (1) and (2): OLS. Column (3): Poisson. SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$..

Heterogeneous productivity effects are mainly driven by team composition effect ▸ Results ▸ Specification

Robustness

Back to Mgr. Classification

- Overall impacts of implementing the system
  - DiD Corrections following Callaway and Sant'Anna (2021), De Chaisemartin and d'Haultfoeuille (2020), and Sun and Abraham (2021). results
- Heterogeneous impacts on teams led by different managers
  - DiD corrections results
  - Alternative manager classification
    - Change the mean leader allowance threshold from the 75% percentile to 50% percentile results
    - Classify managers using manager fixed effects estimated following Abowd et al. (1999) and Lazear et al. (2015). results
    - Use continuous measure for managerial quality results
  - Alternative worker classification results Detail
  - Keep only incumbent workers Results
  - 3 teams with rotations:
    - Drop them and findings the same results. results
    - Test whether replacing a good manager affects team productivity and the likelihood of good workers leaving. results

Manager Classification

GM: better task allocation

- Good workers' workload is 53.7% lower under good managers than under other managers ▸ Specification

VARIABLES	(1)	(2)	(3)	(4)
	IHS(Workload)			
	All Workers	High Performers	Low Performers	New Workers
Supervised by a Good Manager	-0.347***	-0.537***	0.098	1.405*
	(0.123)	(0.129)	(0.279)	(0.783)
Observations	12,962	8,169	2,748	1,920
R-squared	0.522	0.526	0.518	0.687
Worker FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

Notes. Obs. at worker-day level. SE clustered at worker level. $***p < 0.01, **p < 0.05, *p < 0.1$.

- This reduction of workload **mainly comes from menial tasks** ▸ Results by Complexity
  - *Complex tasks*: demand high cognitive effort (e.g. component installation, wiring, spot welding)
  - *Menial tasks*: involve routine physical work (e.g. hoisting, frame installation, painting, cable laying).

GM: better task allocation

- Good workers' workload is 53.7% lower under good managers than under other managers ▸ Specification

VARIABLES	(1)	(2)	(3)	(4)
	IHS(Workload)			
	All Workers	High Performers	Low Performers	New Workers
Supervised by a Good Manager	-0.347***	-0.537***	0.098	1.405*
	(0.123)	(0.129)	(0.279)	(0.783)
Observations	12,962	8,169	2,748	1,920
R-squared	0.522	0.526	0.518	0.687
Worker FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

Notes. Obs. at worker-day level. SE clustered at worker level. $***p < 0.01, **p < 0.05, *p < 0.1$.

- This reduction of workload **mainly comes from menial tasks** ▸ Results by Complexity
  - *Complex tasks*: demand high cognitive effort (e.g. component installation, wiring, spot welding)
  - *Menial tasks*: involve routine physical work (e.g. hoisting, frame installation, painting, cable laying).

Example of complex tasks

- Here the worker is connecting different components with wires according to the design diagram.



Example of menial tasks

- Here the two workers are operating a crane to install the assembled transformer into the cabinet frame.



Less overloading: mistakes ↓ and absence rates ↓

- Good workers **make fewer mistakes** and **have lower absence rates** under good managers

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Making Mistakes				Absent Rate			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a Good Manager	-0.319***	-0.348***	-0.055	-1.030	-0.000	-0.014***	-0.007	0.044**
	(0.066)	(0.075)	(0.186)	(0.675)	(0.005)	(0.003)	(0.009)	(0.018)
Observations	12,962	8,169	2,748	1,920	14,607	6,371	3,220	5,016
R-squared	0.310	0.326	0.340	0.470	0.249	0.113	0.254	0.293
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Obs. at: worker-day level (Col. (1) - (4)); worker-month level (Col. (5) - (8)). SE clustered at worker level. $***p < 0.01, **p < 0.05, *p < 0.1$.

Less self-reported overload

- In the survey:
  - claim: **"feel that my workload is too heavy."**
  - options: *highly agree, agree, disagree, and highly disagree.*
  - A worker is **overload** if they chose *agree* or *highly agree*.
- Good workers are significantly **less likely to feel overburdened** under good managers than under other managers.
  - The effect is −0.741 (0.688 sample mean for control group)



Only for good workers

Less self-reported overload

- In the survey:
  - claim: **"feel that my workload is too heavy."**
  - options: *highly agree, agree, disagree, and highly disagree.*
  - A worker is **overload** if they chose *agree* or *highly agree*.
- Good workers are significantly **less likely to feel overburdened** under good managers than under other managers.
  - The effect is -0.741 (0.688 sample mean for control group)



Only for good workers

Alternative explanations

1. Better payment

   ▸ Details

2. More promotion opportunity

   ▸ Details

3. Less material shortage

   ▸ Details

4. Others: benefits, perception of manager ability, job meaning, and communication quality

   ▸ Details

Alternative explanations

1. Better payment ✗

   ▸ Details

2. More promotion opportunity ✗

   ▸ Details

3. Less material shortage ✗

   ▸ Details

4. Others: benefits, perception of manager ability, job meaning, and communication quality ✗

   ▸ Details

Putting all the evidence together

- Good managers allocate tasks better
  - Good workers receive less menial workload
- GW are less likely to be overloaded under GM
  - Make less mistakes;
  - have lower absence rates;
  - less likely to report overloaded;
- The heterogeneous impacts of the tracking system
  - all managers identify good workers better;
  - only bad managers start overburdening good workers

    $\Longrightarrow$ Good worker retention $\downarrow$, team productivity $\downarrow$

Final remarks

- Information technology alone does not guarantee better performance.
- Managers' ability to protect worker well-being is critical for worker retention and long-run productivity.
- Information technology **amplifies** these managerial capabilities rather than replacing them.

Final remarks

- Information technology alone does not guarantee better performance.
- Managers' ability to protect worker well-being is critical for worker retention and long-run productivity.
- Information technology **amplifies** these managerial capabilities rather than replacing them.

Final remarks

- Information technology alone does not guarantee better performance.
- Managers' ability to protect worker well-being is critical for worker retention and long-run productivity.
- Information technology **amplifies** these managerial capabilities rather than replacing them.


RAHMET
OBRIGADO
GRAZIE
HVALA
谢谢
KIITOS
DANKE
TAK
DANK JE
THANK YOU
SHUKRAAN
ευχαριστώ
GRACIAS
BARKA
MERCI
СПАСИБО
ARIGATO
TAKK
TACK

References I

Abowd, J. M., Kramarz, F., and Margolis, D. N. (1999). High wage workers and high wage firms. *Econometrica*, 67(2):251–333.

Blader, S., Gartenberg, C., and Prat, A. (2020). The contingent effect of management practices. *The Review of Economic Studies*, 87(2):721–749.

Bloom, N., Garicano, L., Sadun, R., and Van Reenen, J. (2014). The distinct effects of information technology and communication technology on firm organization. *Management Science*, 60(12):2859–2885.

Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

De Chaisemartin, C. and d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.

References II

Falk, A. and Kosfeld, M. (2006). The hidden costs of control. *American Economic Review*, 96(5):1611–1630.

Frey, B. S. (1993). Shirking or work morale? the impact of regulating. *European Economic Review*, 37(8):1523–1532.

Kelley, E. M., Lane, G., and Schönholzer, D. (2024). Monitoring in small firms: Experimental evidence from kenyan public transit. *American Economic Review*, 114(10):3119–3160.

Ko, Y. H. and Baek, I. G. (2025). The effect of computer monitoring on employees' productivity in telecommuting arrangements. *Management Science*, 71(1):103–122.

Lazear, E. P., Shaw, K. L., and Stanton, C. T. (2015). The Value of Bosses. *Journal of Labor Economics*, 33(4):823–861.

References III

Pierce, L., Snow, D. C., and McAfee, A. (2015). Cleaning house: The impact of information technology monitoring on employee theft and productivity. *Management Science*, 61(10):2299–2319.

Ranganathan, A. and Benson, A. (2020). A numbers game: Quantification of work, auto-gamification, and worker productivity. *American Sociological Review*, 85(4):573–609.

Staats, B. R., Dai, H., Hofmann, D., and Milkman, K. L. (2017). Motivating process compliance through individual electronic monitoring: An empirical examination of hand hygiene in healthcare. *Management Science*, 63(5):1563–1585.

Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.

Why a model?

Digital tracking gives *all* managers access to richer, worker-specific performance signals, yet the data show:

- Managers still differ sharply in how they match workers to tasks;
- Only some managers show meaningful productivity gains, while others show little improvement or even declines.

A simple model helps clarify:

- How richer worker information *should* influence task assignments?
- How managerial skill changes the value of this information?
- Why digital tracking benefits some teams and not others?

Key ingredients of the model: I

- A production team consists of a manager and a continuum of workers, each either **high-skill** (H) or **low-skill** (L).
- A unit mass of tasks is composed of three types: **Simple** (S), **Physical-intensive** (P), and **Cognitive-intensive** (C).
- The manager assigns each worker one task; team performance is the expected mass of successfully completed tasks.
  - H-workers complete all tasks with certainty.
  - L-workers: always succeed on S, succeed on P with probability $q \in (0, 1)$, and fail all C-tasks.
  - Thus, H-workers have absolute advantages in all tasks, but their comparative advantage lies in C-tasks.

Key ingredients of the model: II

- Managers differ in how effectively they recognize the "right" worker-task match.
  - *Good* managers distinguish P- from C-tasks and understand workers' comparative strengths.
  - *Myopic* managers apply a coarse rule that treats all non-simple tasks as uniformly "difficult".
- **Pre-tracking:** worker types are unobserved, so assignments are effectively random (and managerial skill does not matter).
- **Post-tracking:** worker types become observable, enabling more targeted assignment decisions.

Main theoretical results: I

If tasks arrive and are assigned simultaneously, all teams benefit from tracking, but the productivity gains are larger under good managers.

- Intuitively, all managers improve on random assignment by no longer placing high-skill workers on simple tasks.
- Good managers further direct all H-workers to C-tasks, while myopic managers misallocates part of this capacity toward P-tasks.

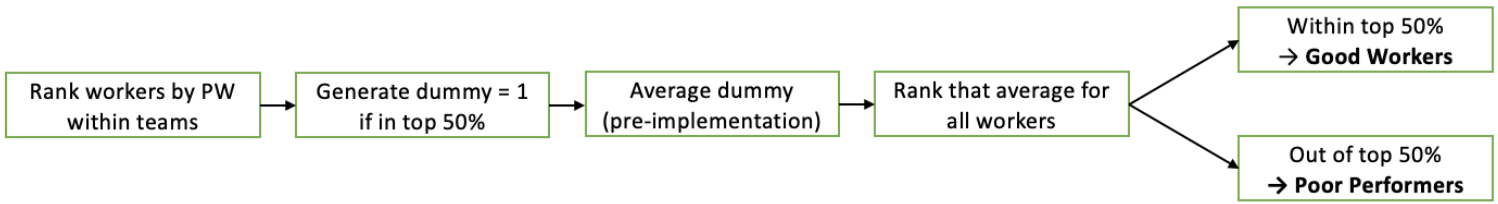
Main theoretical results: II

If tasks arrive and are assigned dynamically, myopic managers can even experience a productivity loss.

- Intuitively, good managers hold back H-workers for C-tasks, but myopic managers follow their coarse rule each period.
- When P-tasks arrive first, they assign H-workers immediately.
- High-skill capacity is depleted early, so most later C-tasks fail.

Define good workers

- Our classification method:
  - Main idea: **workers who often ranks among the top within the team**



- **Good workers:** average ranking **within teams** in the top 50% is within the top 50% among all workers who joined the organization before system implementation
  - **Robustness:** average ranking **within teams** in the **top 25%** is within the top 50% ("generate dummy = 1 if in top 25%" in the second block and other things equal) ▸ Robustness Details

▸ Validation ▸ Back

Manager classification

- We use *leader allowance* to classify team managers:
  - **Leader allowance:** determined by senior management
  - **High-performing managers:** average leader allowance before implementation is higher than the 75% percentile
    - Robustness: 50% percentile/manager fixed effects/continous measure
  - Being led by high-performing managers ↑ output per worker by 13.9% (before implementation) ▸ Table

- Cannot be significantly predicted by demographic variables and survey measured beliefs and attitudes ▸ Graphs ▸ Back

Manager classification

- We use *leader allowance* to classify team managers:
  - **Leader allowance:** determined by senior management
  - **High-performing managers:** average leader allowance before implementation is higher than the 75% percentile
    - Robustness: 50% percentile/manager fixed effects/continous measure
  - Being led by high-performing managers ↑ output per worker by 13.9% (before implementation) ▸ Table

- Cannot be significantly predicted by demographic variables and survey measured beliefs and attitudes ▸ Graphs ▸ Back

Classifying teams (time-invariant)

- Team classification:
  - **Teams led by good managers:** whether the teams had at least one good manager before implementation (16 teams out of 38)
  - In our sample, **very scarce managers' rotation**
  - After the implementation
  - 13 out of 16 continued to have good managers
    - For robustness, we dropped the 3 switching teams ▸ Robustness Details
  - all 22 teams without good manager before still had no good manager

▸ Back

Summary statistics

Variable	Obs. #	Mean	Standard Deviation	10%	25%	50%	75%	90%
Panel A. Output Variables								
Total Output (thousand yuan)	1029	190.23	157.59	52.70	84.34	137.70	244.85	411.18
Output per Worker (thousand yuan)	1029	10.93	3.05	7.09	8.79	10.86	12.89	14.71
Panel B. Human Resource Variables								
Worker No.	1029	16.93	11.69	5	8	13	23	35
Worker per Manager	1029	8.75	5.62	3.41	5.00	7.50	11.00	15.00
Manager No.	1029	1.84	1.18	1	1	1	2	3
New Worker No.	1029	1.33	2.57	0	0	0	2	4
Good Workers Transferred In	1029	0.93	2.52	0	0	0	0	3
Poor Performers Transferred In	1029	0.52	1.48	0	0	0	0	2
Good Workers Transferred Out	1029	0.93	2.51	0	0	0	0	3
Poor Performers Transferred Out	1029	0.52	1.49	0	0	0	0	2
Turnover Rate for Good Workers	1029	0.00	0.02	0	0	0	0	0
Turnover Rate for Poor Performers	1029	0.01	0.03	0	0	0	0	0
Panel C. Demographic Characteristics								
Manager %Female	1029	0.16	0.35	0	0	0	0	1
Worker %Female	1029	0.18	0.27	0	0	0	0.26	0.67
Manager %Ethnic Han	1029	0.95	0.20	1	1	1	1	1
Worker %Ethnic Han	1029	0.98	0.05	0.92	1	1	1	1
Manager %CCP/CLY Member	1029	0.40	0.42	0	0	0.33	1	1
Worker %CCP/CLY Member	1029	0.26	0.16	0.07	0.15	0.24	0.33	0.46
Manager %College	1029	0.20	0.34	0	0	0	0.33	1
Worker %College	1029	0.21	0.16	0	0.08	0.19	0.30	0.43
Average Manager Age	1029	35.23	7.19	25.20	30.33	34.75	40.00	45.00
Average Worker Age	1029	33.16	5.83	27.04	29.07	31.94	35.55	42.12
Average Manager Experience	1029	8.39	6.36	1.50	3	7.50	12	16
Average Worker Experience	1029	4.23	4.38	0.66	1.46	2.65	4.84	11.17
Panel D. Managerial Quality Variable								
Team with Good Managers	1029	0.41	0.49	0	0	0	1	1

Notes. Each obs. is a team $\times$ bimonthly period.

back

Good managers and productivity

Table: Managerial Skill and Productivity

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	IHS(Output per Worker)			Output per Worker		
	Pooled	Before	After	Pooled	Before	After
High-performing Manager	0.120*	0.139***	0.111	1,490.638	1,143.668***	1,665.356
	(0.066)	(0.033)	(0.080)	(947.362)	(297.851)	(1,214.953)
Observations	933	268	659	933	268	659
R-squared	0.690	0.793	0.715	0.683	0.831	0.689
Team Controls	Yes	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Obs. at team-bimontly period level. SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$..

back

Example of task adjustment measures

- For example, a worker was assigned
  - 10 tasks/3 types on day $t-1$, 11 tasks/4 types on day $t \Longrightarrow$ $D1=1, D2=1$;
  - 10 tasks/3 types on day $t-1$, 10 tasks/4 types on day $t \Longrightarrow$ $D1=0, D2=1$;
  - 10 tasks/3 types on day $t-1$, 11 tasks/3 types on day $t \Longrightarrow$ $D1=1, D2=0$;
  - 10 tasks/3 types on day $t-1$, 10 tasks/3 types on day $t \Longrightarrow$ $D1=0, D2=0$;

back

Relative to other managers, high-performing managers:

- **More experimentally for new workers:**
  - 20.4% ↑ to change no. of tasks, 27.4% ↑ to change no. of task types
- **More stable for experienced high-performing workers:**
  - 5.8% ↓ to change no. of tasks, 8.9% ↓ to change no. of task types

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Change No. of Tasks				Change No. of Task Types			
	All Workers	High Performers	Poor Performers	New Workers	All Workers	High Performers	Poor Performers	New Workers
Supervised by a Good Manager	-0.055***	-0.058***	-0.055	0.204***	-0.058*	-0.089**	0.044	0.274***
	(0.017)	(0.018)	(0.056)	(0.071)	(0.034)	(0.041)	(0.059)	(0.088)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.178	0.212	0.258	0.316	0.301	0.331	0.346	0.400
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Obs. at worker-day level. SE clustered at worker level. $***p < 0.01, **p < 0.05, *p < 0.1$.

back

Balance tests

- **ever-treated and never-treated (levels):** ever-treated teams have higher %age<30 and higher %experience<5 yrs $\Longrightarrow$ **we control for these**
- **ever-treated and never-treated (diffs):** no significant difference
- **ologit of implement order on team characteristics:** no characteristic significantly predict order

Balance tests results

Variable	(1) mean of never takers	(2) β of Ever takers(Levels)	(3) β of Ever takers(First-Diff)	(4) Timing of implementation
Manager %Female	0.0495	0.2079	0.0006	4.2171
		(0.1856)	(0.0718)	(5.1605)
Worker %Female	0.0745	0.0532	-0.0024	6.3740
		(0.1322)	(0.0518)	(7.5534)
Manager %Ethnic Han	0.9904	-0.0022	0.0000	0.0000
		(0.0118)	(0.0000)	(0.0000)
Worker %Ethnic Han	0.9769	-0.0031	-0.0003	-20.6586
		(0.0120)	(0.0036)	(37.1164)
Manager %CCP/CLY Member	0.4692	-0.0969	-0.1707	-2.4894
		(0.1986)	(0.1417)	(2.0261)
Worker %CCP/CLY Member	0.1839	0.1487	-0.0214	-3.5355
		(0.1257)	(0.0498)	(4.7114)
Manager %College	0.1370	0.1815	0.0166	1.9138
		(0.1682)	(0.0669)	(3.7574)
Worker %College	0.1787	0.0574	-0.0037	11.3304
		(0.0930)	(0.0395)	(5.5094)
Manager %Age<35	0.5139	0.1733	-0.0850	5.1649
		(0.2235)	(0.1163)	(2.0846)
Manager %Age≥35	0.4861			
Worker %Age<30	0.2964	0.3239**	-0.0590	1.4125
		(0.1423)	(0.0674)	(2.9796)
Worker %Age≥30	0.6476			
Manager %Experience<5	0.3811	0.1585	-0.0046	-4.1609
		(0.2062)	(0.1305)	(2.1071)
Manager %Experience≥5	0.6189			
Worker %Experience<5	0.6647	0.3097**	0.0044	4.6497
		(0.1279)	(0.0340)	(4.8733)
Worker %Experience≥5	0.2582			

Notes. Obs at team bimonthly period level.

back

Treatment effects decomposition: results

Table: Treatment Effects Holding Team Composition Fixed

VARIABLES	(1)	(2)	(3)	(4)
	CSPP		IHS(CSPP)	
Implemented	475.722***	334.174**	0.416*	0.308
	(146.669)	(163.323)	(0.214)	(0.226)
Implemented × High-Performing Manager		411.168		0.336
		(318.323)		(0.389)
Observations	933	866	933	866
R-squared	0.784	0.719	0.886	0.458
Team Controls	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes
Period FE	Yes		Yes	
Group × Period FE		Yes		Yes

Notes. Obs. at team-bimonthly period level. SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$.

Back

Robustness: 50% criteria for mgr. (output vars)

VARIABLES	(1) IHS(Output)	(2) IHS(Output per Worker)
Implemented	0.425***	-0.059*
	(0.150)	(0.034)
Implemented * with Good Manager	-0.149	0.156**
	(0.207)	(0.065)
Observations	955	955
R-squared	0.823	0.690
Control Mean	12.05	9.82
Team Controls	Yes	Yes
Team FE	Yes	Yes
Group*Period FE	Yes	Yes

Notes. Obs. at team-bimontly period level. SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$..

▸ Back to Robustness ▸ Specification

Robustness: 50% criteria for mgr. (personnel vars)

VARIABLES	(1) Workers	(2) Managers	(3) IHS(worker per Manager)	(4) New Workers	(5) High Performers Transferred In	(6) **High Performers Transferred Out**	(7) Low Performers Transferred In	(8) Low Performers Transferred Out	(9) High Performers Turnover	(10) Low Performers Turnover
Implemented	0.485***	0.360***	0.172	1.110*	1.148	**1.352**	-0.231	0.206	-0.003	-0.005
	(0.147)	(0.125)	(0.125)	(0.575)	(0.850)	**(1.111)**	(1.021)	(1.295)	(0.003)	(0.006)
Implemented * with Good Manager	-0.280	-0.262	-0.041	-0.728	-0.915	**-2.232***	0.387	-0.729	0.008	0.010
	(0.176)	(0.193)	(0.151)	(0.593)	(0.882)	**(1.215)**	(1.202)	(1.417)	(0.005)	(0.009)
Observations	955	955	951	910	572	653	693	698	955	955
R-squared			0.558						0.173	0.290
Control Mean	11.79	1.42	2.69	1.55	0.42	0.34	0.17	0.16	0.01	0.02
Team Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Group*Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Column (3) use OLS and others use Poisson. Obs. at team-bimontly period level.

SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$..

▸ Back to Robustness ▸ Specification

Robustness: top 25% criteria for workers (output vars)

VARIABLES	(1) IHS(Output)	(2) IHS(Output per Worker)
Implemented	0.367***	-0.063**
	(0.121)	(0.030)
Implemented * with Good Manager	-0.018	0.179**
	(0.212)	(0.066)
Observations	955	955
R-squared	0.819	0.697
Control Mean	12.28	9.83
Team Controls	Yes	Yes
Team FE	Yes	Yes
Group*Period FE	Yes	Yes

Notes. Obs. at team-bimontly period level. SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$..

▸ Back to Robustness ▸ Specification

Robustness: top 25% criteria for worker (personnel vars)

VARIABLES	(1) Workers	(2) Managers	(3) IHS(worker per Manager)	(4) New Workers	(5) High Performers Transferred In	(6) **High Performers Transferred Out**	(7) Low Performers Transferred In	(8) Low Performers Transferred Out	(9) High Performers Turnover	(10) Low Performers Turnover
Implemented	0.424***	0.208**	0.239**	0.738**	0.848	0.902	-0.228	0.203	-0.003	-0.001
	(0.088)	(0.100)	(0.110)	(0.295)	(0.758)	(0.687)	(1.089)	(0.662)	(0.002)	(0.005)
Implemented * with Good Manager	-0.138	-0.088	-0.063	-0.130	-0.659	-1.486*	0.079	-0.517	0.007*	0.008
	(0.149)	(0.189)	(0.140)	(0.369)	(0.815)	(0.814)	(1.230)	(1.003)	(0.004)	(0.010)
Observations	955	955	955	895	575	640	603	673	955	955
R-squared			0.565						0.179	0.302
Control Mean	14.30	1.62	2.79	1.63	0.49	0.44	0.17	0.14	0.01	0.02
Team Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Group*Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Column (3) use OLS and others use Poisson.Obs. at team-bimontly period level.

SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$..

▸ Back to Robustness ▸ Specification

Robustness: continuous measure for managerial quality

- Why binary classification?
  - Theoretically, we care about distinguishing **low vs. high**, not marginal differences
  - It helps **visualize the dynamic differences** between the two types

- Our results still hold using continuous measure:
  - Managerial quality: the team's **maximum pre-implementation leader allowance**

Back to Robustness Tables

Robustness: continuous measure for managerial quality

- Why binary classification?
  - Theoretically, we care about distinguishing **low vs. high**, not marginal differences
  - It helps **visualize the dynamic differences** between the two types

- Our results still hold using continuous measure:
  - Managerial quality: the team's **maximum pre-implementation leader allowance**

▸ Back to Robustness ▸ Tables

Robustness: continuous results

Variable	(1) IHS(Output)	(2) IHS(Output per Worker)	(3) High Performers Transferred Out
Implemented	0.2814	-0.0498	0.6895*
	(0.2594)	(0.0631)	(0.4090)
Implemented × Managerial Quality	0.1831	0.2057*	-2.4900***
	(0.5866)	(0.1095)	(0.6955)
Observations	955	955	642
R-squared	0.8189	0.6951	
Control Mean	12.28	9.83	0.43
Team Controls	Yes	Yes	Yes
Team FE	Yes	Yes	Yes
Group*Period FE	Yes	Yes	Yes

Notes. Obs. at team-bimontly period level. SE clustered at the team level. $***p < 0.01, **p < 0.05, *p < 0.1$.

▸ Back to Robustness ▸ Back to Definition ▸ Specification

Robustness: no switching teams (output vars)

VARIABLES	(1) IHS(Output)	(2) IHS(Output per Worker)
Implemented	0.354***	-0.062**
	(0.123)	(0.028)
Implemented * with Good Manager	-0.127	0.195**
	(0.221)	(0.088)
Observations	895	895
R-squared	0.808	0.691
Control Mean	12.28	9.83
Team Controls	Yes	Yes
Team FE	Yes	Yes
Group*Period FE	Yes	Yes

Notes. Obs. at team-bimontly period level. SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$..

▸ back ▸ Specification

Robustness: no switching teams (personnel vars)

VARIABLES	(1) Workers	(2) Managers	(3) IHS(worker per Manager)	(4) New Workers	(5) High Performers Transferred In	(6) **High Performers Transferred Out**	(7) Low Performers Transferred In	(8) Low Performers Transferred Out	(9) High Performers Turnover	(10) Low Performers Turnover
Implemented	0.397***	0.189**	0.241**	0.779***	0.756	**1.200****	-0.074	-0.092	-0.002	-0.004
	(0.080)	(0.096)	(0.106)	(0.299)	(0.800)	**(0.529)**	(1.225)	(0.711)	(0.002)	(0.005)
Implemented * with Good Manager	-0.294**	-0.258	-0.051	-0.049	-0.807	**-1.698*****	0.055	-0.299	0.009	0.009
	(0.144)	(0.185)	(0.150)	(0.380)	(0.843)	**(0.605)**	(1.351)	(0.928)	(0.006)	(0.010)
Observations	895	895	891	840	552	591	633	647	895	895
R-squared			0.561						0.188	0.308
Control Mean	14.30	1.62	2.81	1.63	0.49	0.43	0.17	0.16	0.01	0.02
Team Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Group*Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Column (3) use OLS and others use Poisson. Obs. at team-bimontly period level.

SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$..

▸ back ▸ Specification

Robustness: the effect of losing a good manager

VARIABLES	(1) IHS(Output per Worker)	(2) High Performers Transferred Out
After Losing Good Manager	-0.218***	0.908
	(0.051)	(1.345)
Observations	564	296
R-squared	0.746	
Team Controls	Yes	Yes
Team FE	Yes	Yes
Period FE	Yes	Yes

Notes. Obs. at team-bimonthly period level. SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$.

back

Workload allocation (alternative workload measure)

- To check robustness, we alternatively define workload as the total **real task duration**.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	ihs(Workload)				Workload			
	All Workers	Good Workers	Poor Performers	New Workers	All Workers	Good Workers	Poor Performers	New Workers
Supervised by a Good Manager	-0.309	-0.519**	0.475	0.914	-835.023*	-1,174.301*	252.524	-41.546
	(0.208)	(0.223)	(0.416)	(0.758)	(500.346)	(634.243)	(271.072)	(963.429)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.443	0.454	0.449	0.590	0.157	0.161	0.292	0.320
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Obs. at worker-day level. SE clustered at worker level. $***p < 0.01, **p < 0.05, *p < 0.1$.

▸ Back ▸ Specification

Workload allocation by complexity

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	IHS(Complex Workload)				IHS(Menial Workload)			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-Performing Manager	-0.041	-0.123	0.310	2.148***	-0.340	-0.501**	0.254	2.225
	(0.148)	(0.197)	(0.220)	(0.687)	(0.211)	(0.241)	(0.289)	(1.339)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.740	0.753	0.707	0.818	0.673	0.667	0.681	0.806
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Variable	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
	Complex Task Number				Menial Task Number			
	All Workers	High Performers	Low Performers	New Workers	All Workers	High Performers	Low Performers	New Workers
Supervised by a High-Performing Manager	-0.484	-0.716	0.026	5.213**	-6.657***	-6.258**	-6.570	-1.925
	(0.500)	(0.785)	(0.335)	(2.213)	(2.204)	(2.440)	(5.473)	(3.959)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.406	0.416	0.437	0.542	0.512	0.510	0.559	0.695
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Obs. at worker-day level. SE clustered at the worker level. $***p < 0.01, **p < 0.05, *p < 0.1$.

▸ Back ▸ Specification

Alternatives: payment and promotion

- If hold, we expect the coefficients for the interact term significantly positive for good workers.

VARIABLES	(1)	(2)	(3)	(4)
	IHS(Performance Wage)		Promotion	
	Good Workers	Poor Performers	Good Workers	Poor Performers
Implemented	0.028	0.154***	1.233	0.642
	(0.031)	(0.046)	(0.851)	(0.487)
Implemented * Good Manager	0.057	0.230*	-2.045	0.841
	(0.057)	(0.119)	(2.401)	(0.770)
Good Manager	-0.031	-0.218**	2.267	-0.744
	(0.038)	(0.103)	(2.764)	(0.796)
Observations	17,961	8,473	17,975	8,479
R-squared	0.464	0.473	0.048	0.111
Worker FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

Notes. Obs. at worker-month level. SE clustered at worker level, $***p < 0.01, **p < 0.05, *p < 0.1$.

▸ Back to Mech. List ▸ Specification

Alternatives: payment and promotion

- In our survey, we have a question:
  - Show the claim: **"I am not fairly treated by my manager"**
  - Workers chose among *highly agree(4)/agree(3)/disagree(2)/highly disagree(1)*
- We regress the response (numerical 1-4) on *dummy(good worker)* and *dummy(good worker)*×*dummy(good mgr)*, controlling team FE:
  - We expect the coefficient for the interact term to be **significantly negative**
  - Actually it is positive $\Longrightarrow$ this cannot explain our results

Back to Mech. List Results

Alternatives: payment and promotion

- In our survey, we have a question:
  - Show the claim: **"I am not fairly treated by my manager"**
  - Workers chose among *highly agree(4)/agree(3)/disagree(2)/highly disagree(1)*
- We regress the response (numerical 1-4) on *dummy(good worker)* and *dummy(good worker)*×*dummy(good mgr)*, controlling team FE:
  - We expect the coefficient for the interact term to be **significantly negative**
  - Actually it is positive $\Longrightarrow$ this cannot explain our results

▸ Back to Mech. List ▸ Results

Alternatives: fairness

- Raw response: 1 (highly disagree), 2 (disagree), 3 (agree), and 4 (highly agree)
- Agree dum.: $= 1$ if chose agree/highly agree, otherwise $= 0$.

VARIABLES	(1) Raw Response	(2) Agree Dum.
Good Worker	0.141	0.037
	(0.137)	(0.056)
Good Manager * Good Worker	0.398*	0.214**
	(0.214)	(0.085)
Observations	578	578
R-squared	0.098	0.077
Team FE	Yes	Yes

Notes. Obs. at worker level. SE clustered at team level, $***p < 0.01, **p < 0.05, *p < 0.1$.

Back

Alternatives: adoption and material shortage

- Adoption:
  - Good managers $\rightarrow$ better adoption $\rightarrow$ higher productivity
  - We find only better adoption for good workers
  - Poor Performers achieve even higher productivity gain ▸ Decomposition
  - **cannot explain the productivity gain for only poor performers** (see next page)
- Material Shortage: no significant difference

▸ Back to Mech. List ▸ Specification

Alternatives: adoption and material shortage

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No Auto-termination				No Shortage			
	All Workers	Good Workers	Poor Performers	New Workers	All Workers	Good Workers	Poor Performers	New Workers
Supervised by a Good Manager	0.051***	0.059***	0.016	0.213	0.036	0.034	0.039	-0.105
	(0.019)	(0.022)	(0.033)	(0.161)	(0.039)	(0.042)	(0.126)	(0.080)
Observations	12,962	8,169	2,748	1,920	12,962	8,169	2,748	1,920
R-squared	0.439	0.435	0.434	0.542	0.379	0.385	0.460	0.514
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors are clustered at the worker level. $***p < 0.01, **p < 0.05, *p < 0.1$.

▸ Back to Mech. List ▸ Specification

Decomposing productivity gains

VARIABLES	(1)	(2)	(3)	(4)
	Output per Worker (Good)		Output per Worker (Poor)	
	original	IHS	original	IHS
Implemented	639.703	0.064	85.197	0.039
	(501.300)	(0.042)	(406.136)	(0.058)
Implemented * with Good Manager	1,264.663	0.063	2,659.784**	0.249**
	(1,153.568)	(0.070)	(1,225.080)	(0.106)
Observations	950	950	903	903
R-squared	0.702	0.699	0.698	0.658
Team Controls	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes
Group*Period FE	Yes	Yes	Yes	Yes

Notes. Obs. at team-bimontly period level. SE clustered at team level. $***p < 0.01, **p < 0.05, *p < 0.1$.

Back

Other alternative mechanisms

	(1)	(2)	(3)	(4)
	Benefits and Opportunities		Perceived Manager Ability	
VARIABLES	Promotion Chance	Benefits Provided	Good Scheduling	Good Cooperation
Good Worker	-0.182	-0.116	-0.154	-0.058
	(0.144)	(0.104)	(0.111)	(0.081)
Good Manager * Good Worker	-0.060	-0.032	-0.028	-0.139
	(0.210)	(0.145)	(0.168)	(0.144)
Observations	578	578	578	578
R-squared	0.084	0.086	0.048	0.050
Team FE	Yes	Yes	Yes	Yes
	(5)	(6)	(7)	(8)
	Job Meaning		Communication Quality	
VARIABLES	Importance for Company	Importance for Society	Communication Times	Communication Smoothness
Good Worker	-0.059	0.044	0.765	-0.063
	(0.091)	(0.087)	(0.849)	(0.124)
Good Manager * Good Worker	0.191	-0.009	-0.625	-0.253
	(0.180)	(0.171)	(1.336)	(0.168)
Observations	578	578	578	578
R-squared	0.093	0.073	0.087	0.103
Team FE	Yes	Yes	Yes	Yes

Notes. Obs. at worker level. SE clustered at team level. $***p < 0.01, **p < 0.05, *p < 0.1$.

▸ Back to Mech. List ▸ Survey Questions

Survey questions

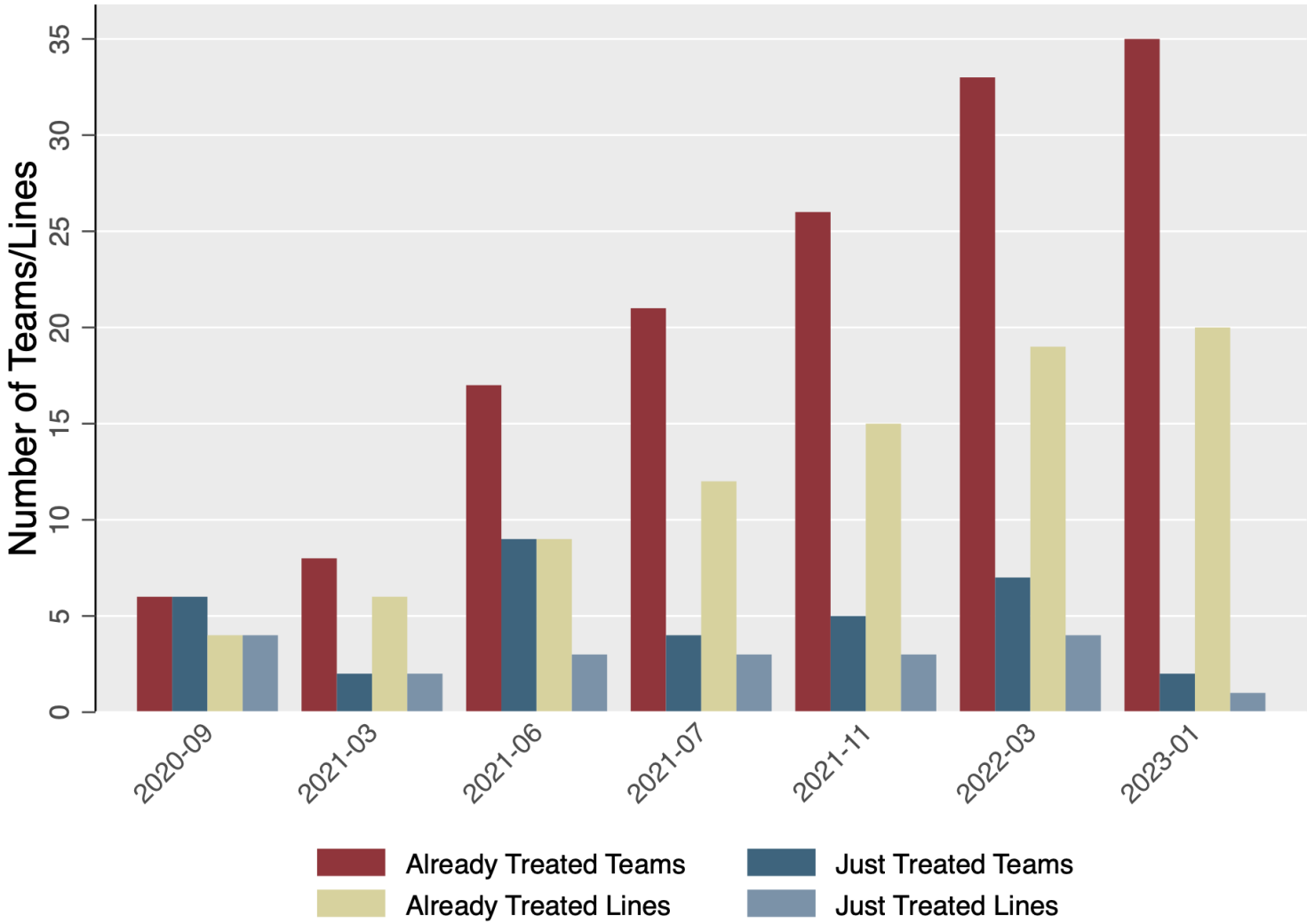
Please choose your attitude towards each of these claims: (highly disagree/disagree/agree/highly agree)

- My promotion opportunity is limited
- I am not satisfied with the benefits issued by the company
- The team manager's organization and scheduling are reasonable
- I think the cooperation in my team is good
- The work I do is important to the company
- The work I do is important to society
- My communication with the team manager is smooth

Comm. Times: How many times do you communicate with your team manager every day?

Back to Results

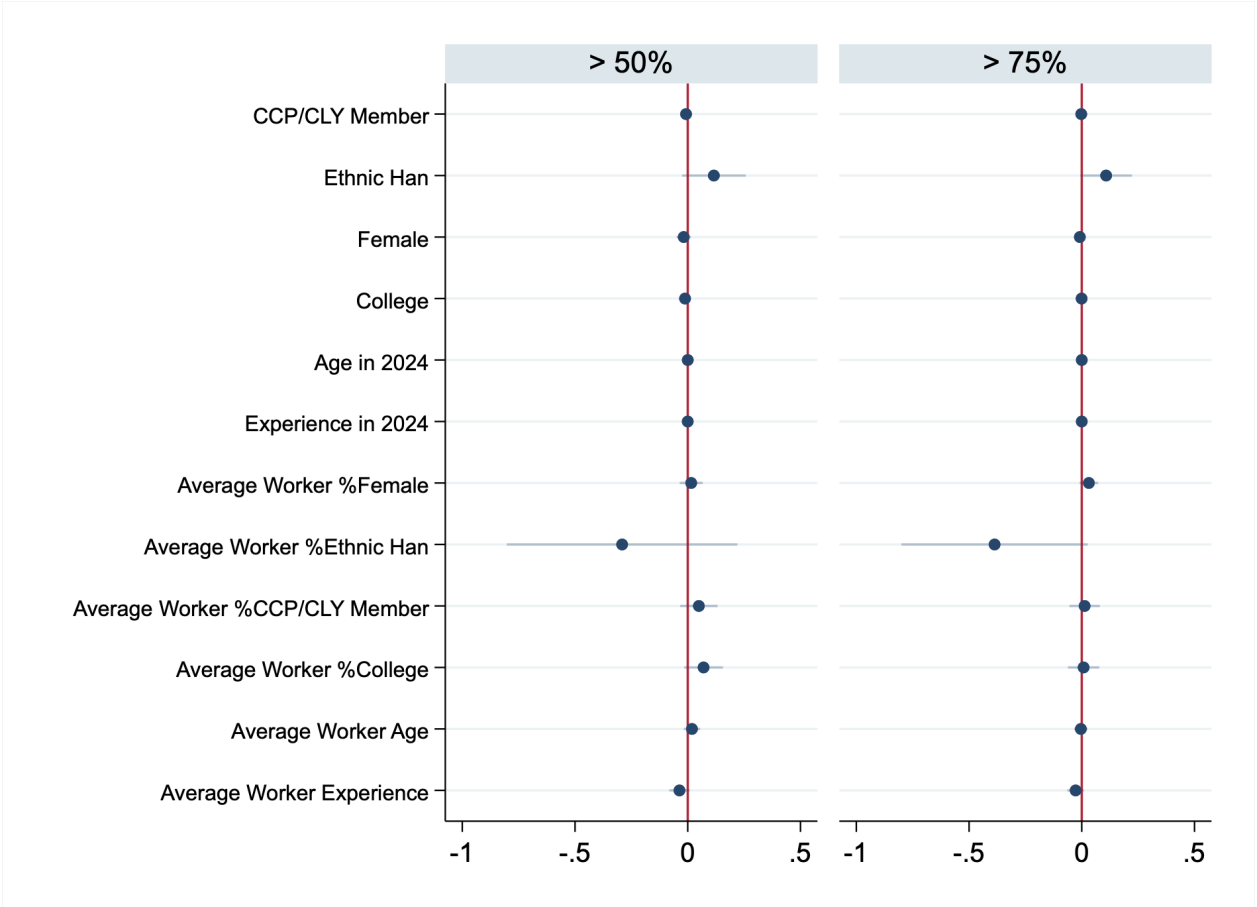
Staggered implementation



Back to Context 35

What factors predict whether a manager is good?

Self + Workers

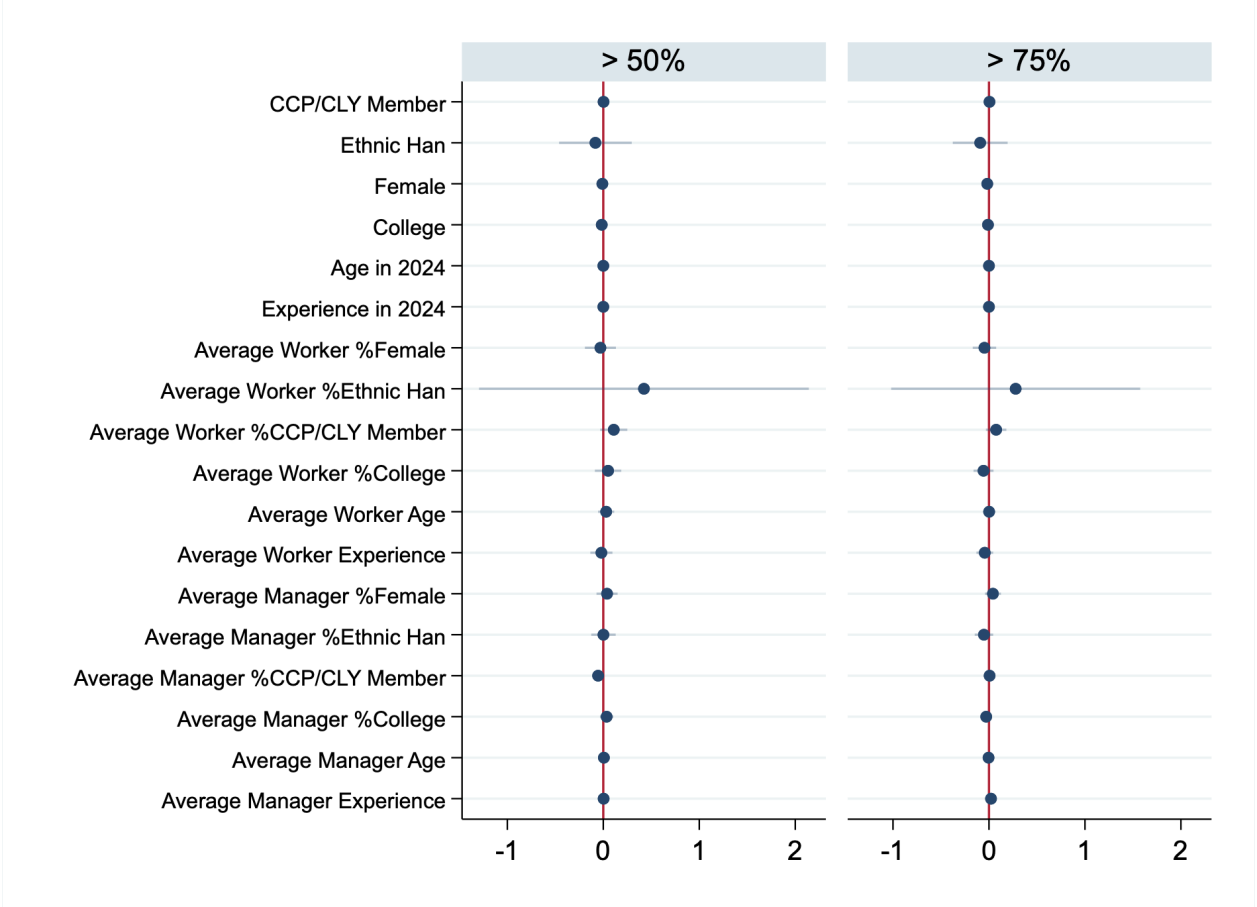


Obs No.: 144 for both

Back

What factors predict whether a manager is good?

Self + Workers + Other Mgrs

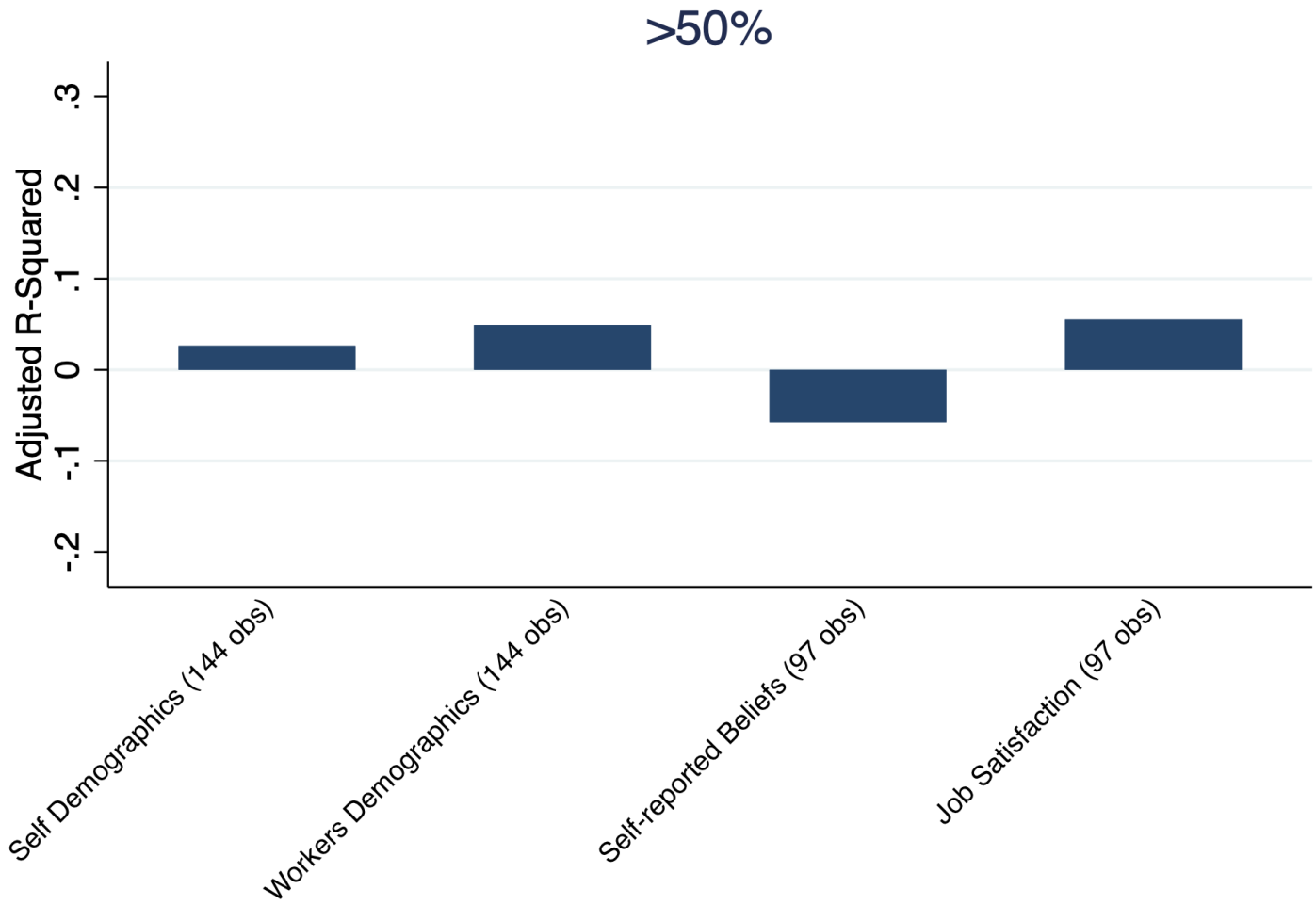


Obs No.: 67 for both

Back

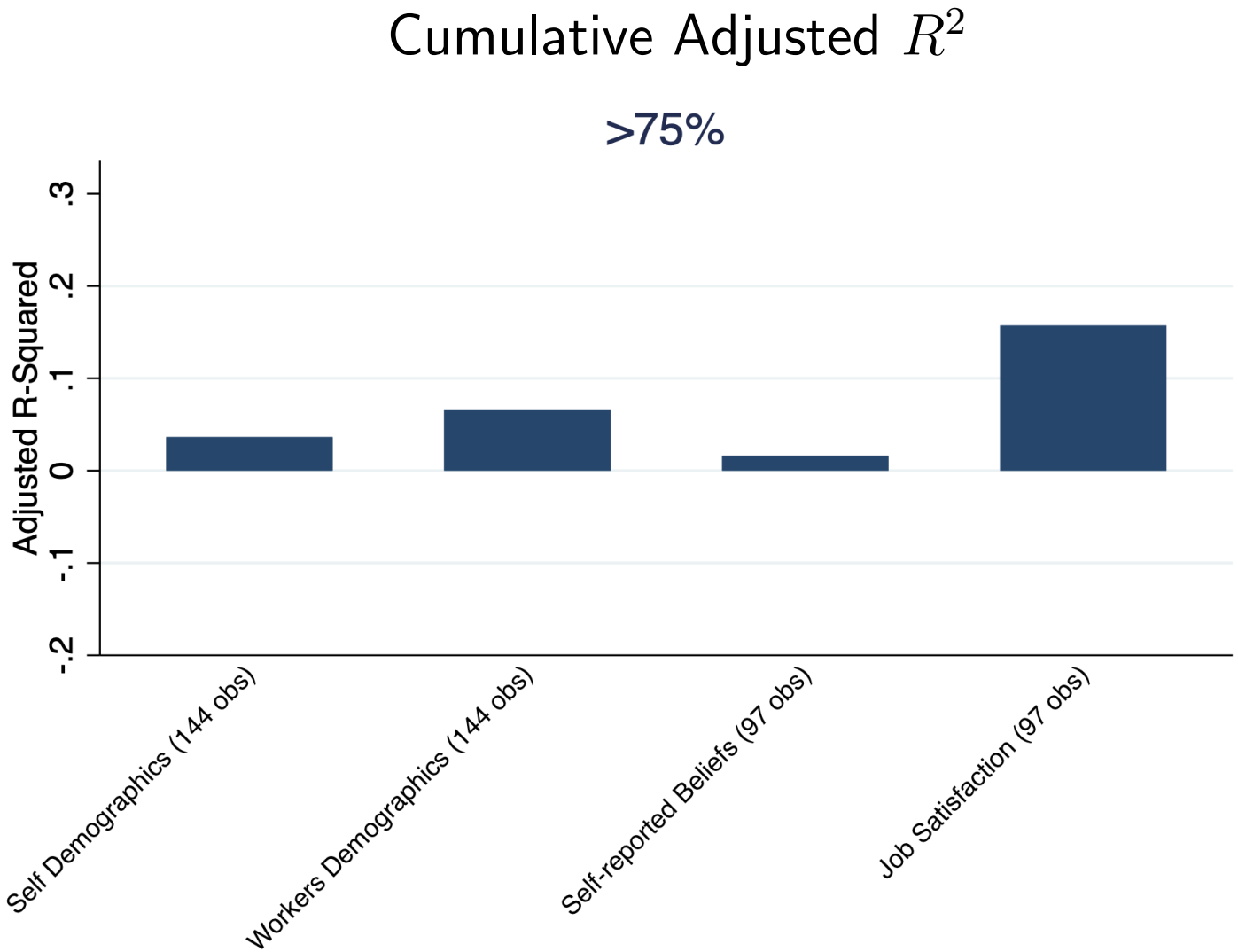
What factors predict whether a manager is good?

Cumulative Adjusted R^2



Back

What factors predict whether a manager is good?



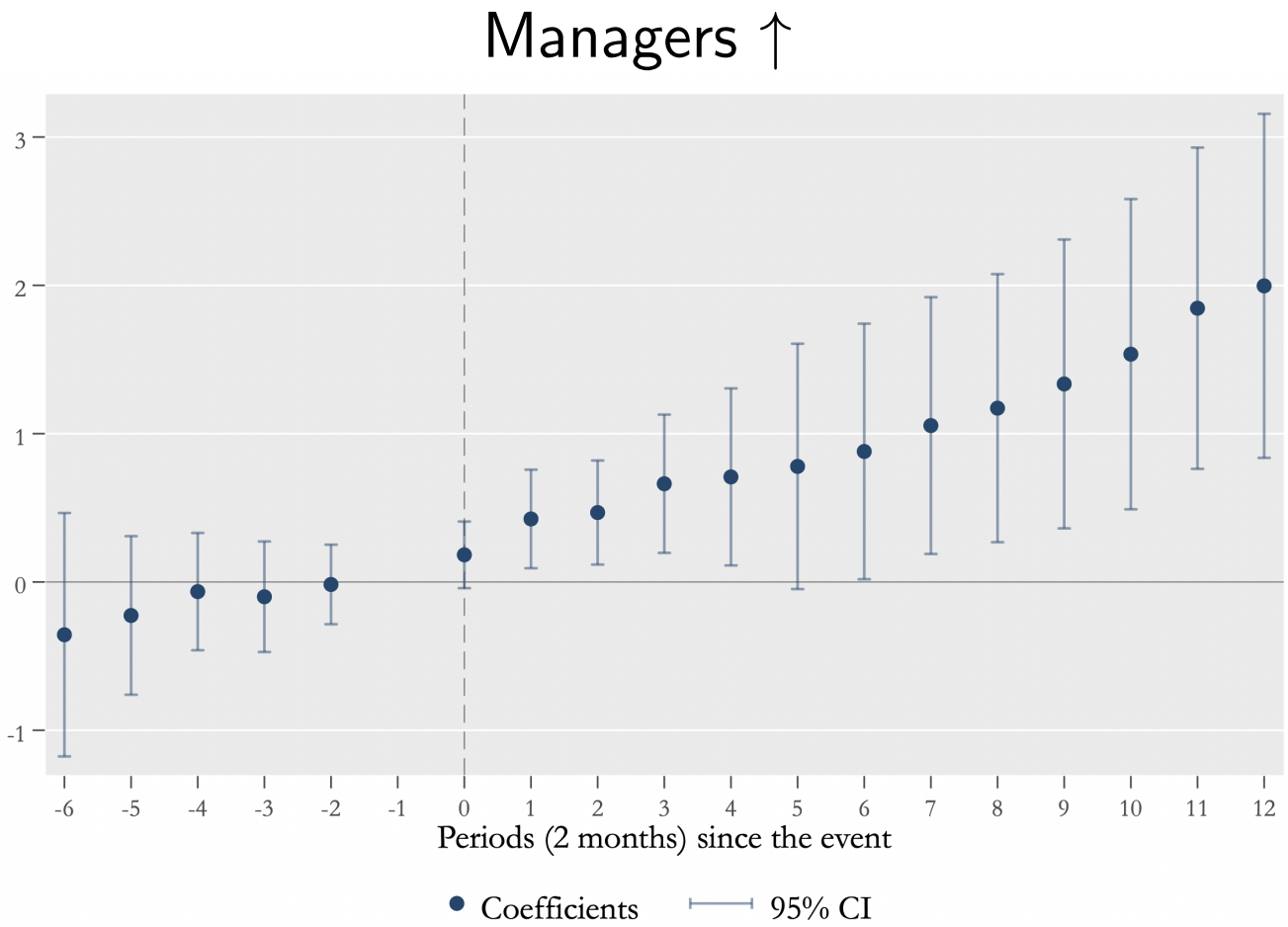
Back

Characterizing manager types



Back

Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

Back

Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

Back

Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back

Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back

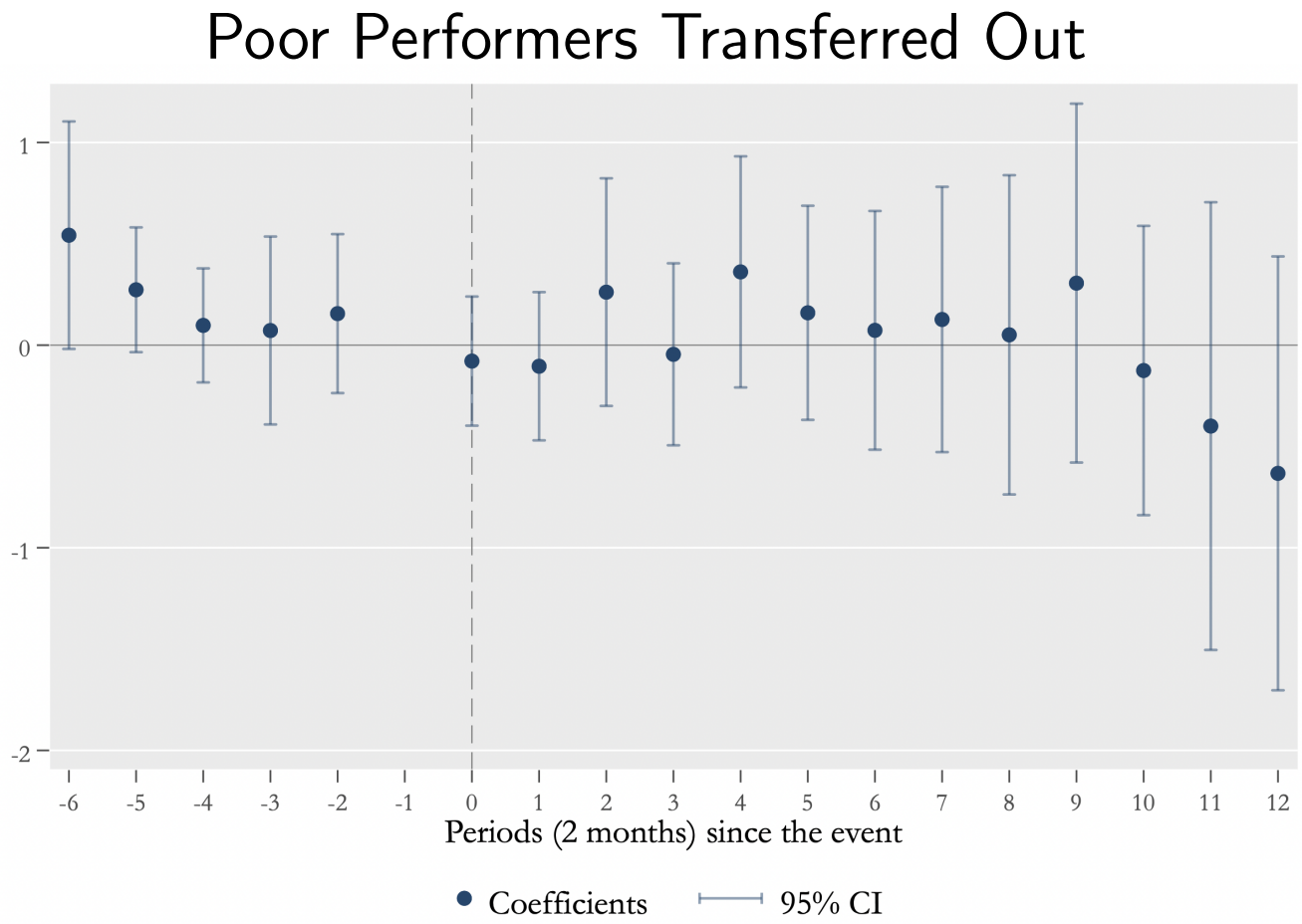
Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back

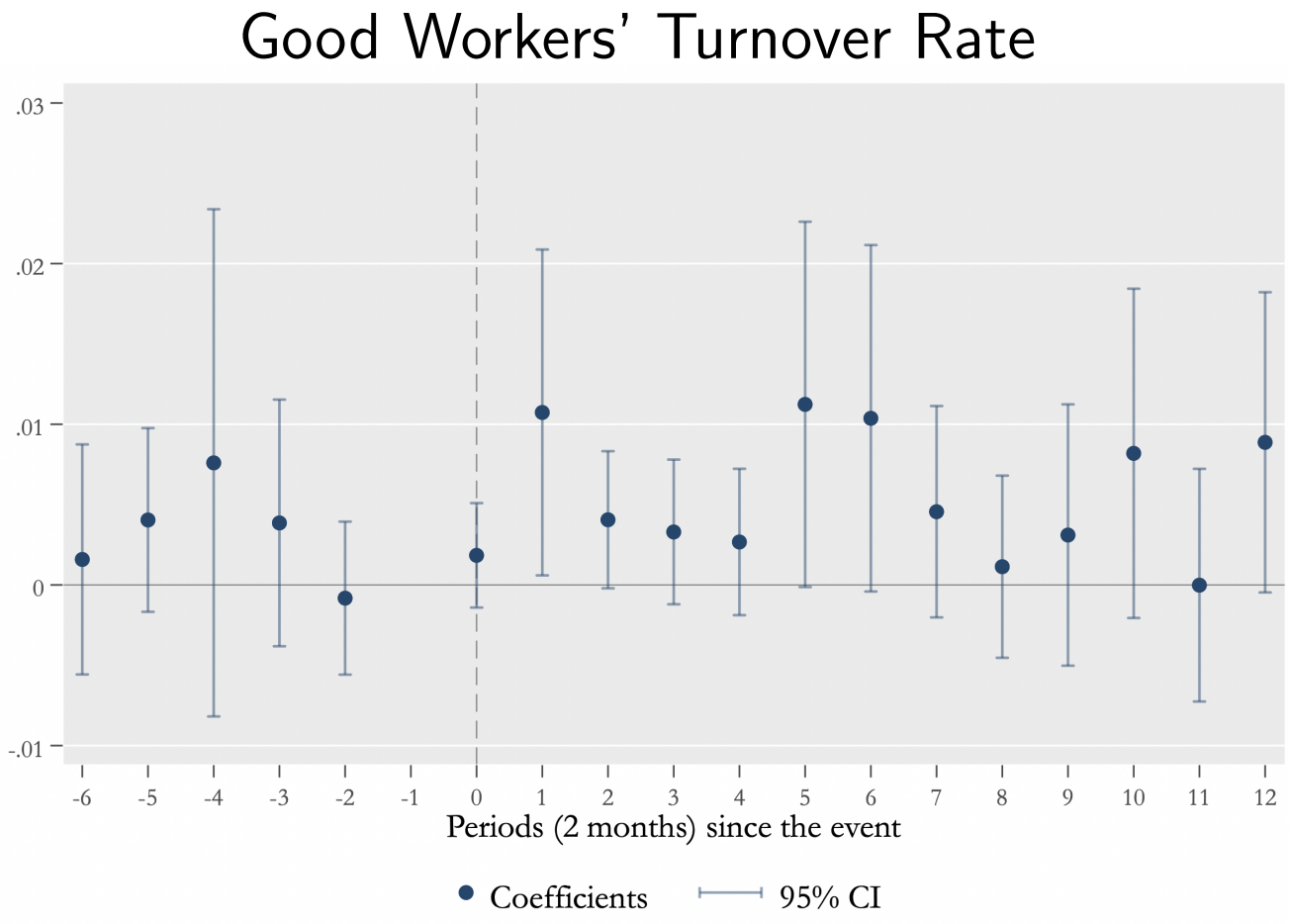
Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back

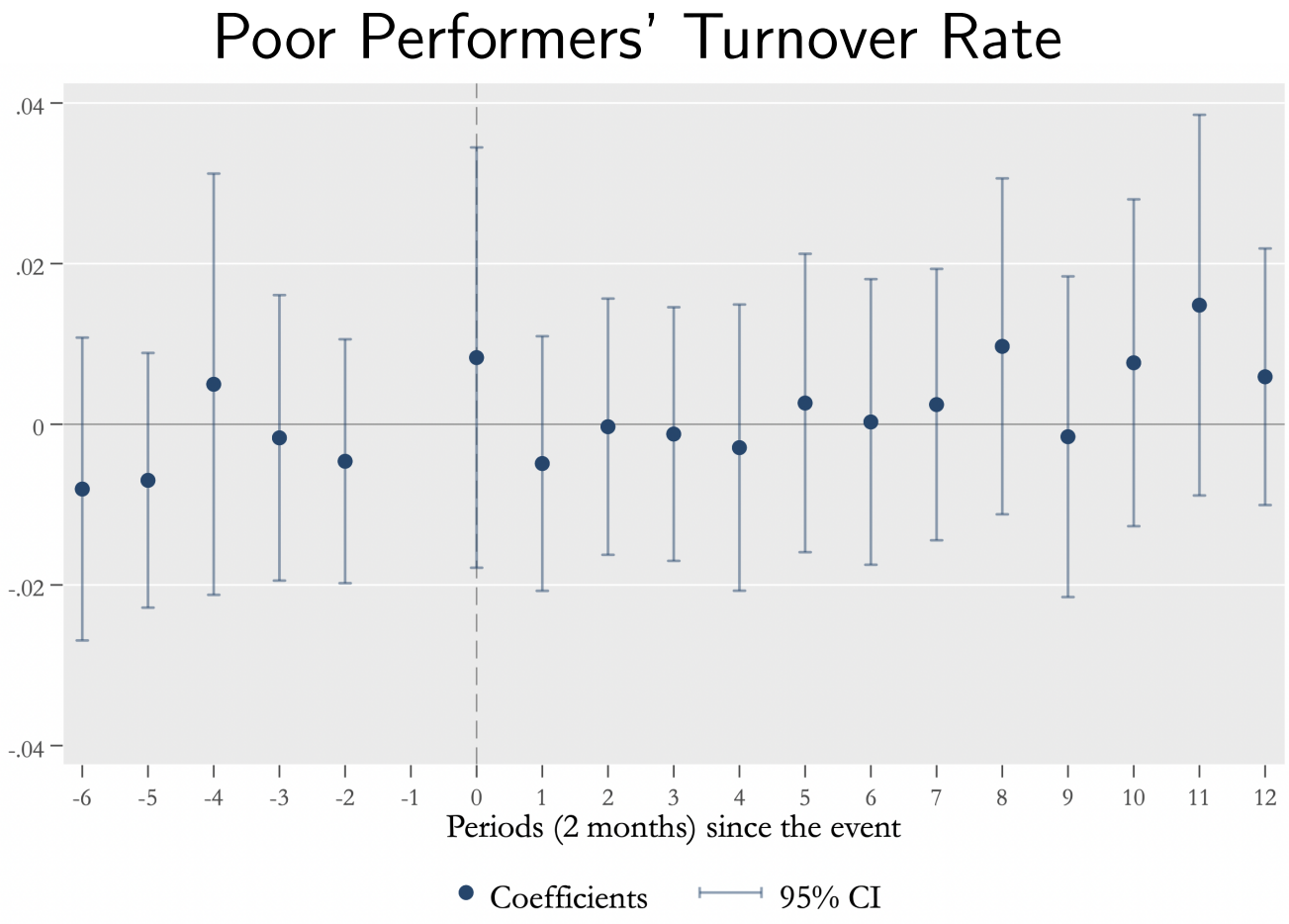
Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

Back

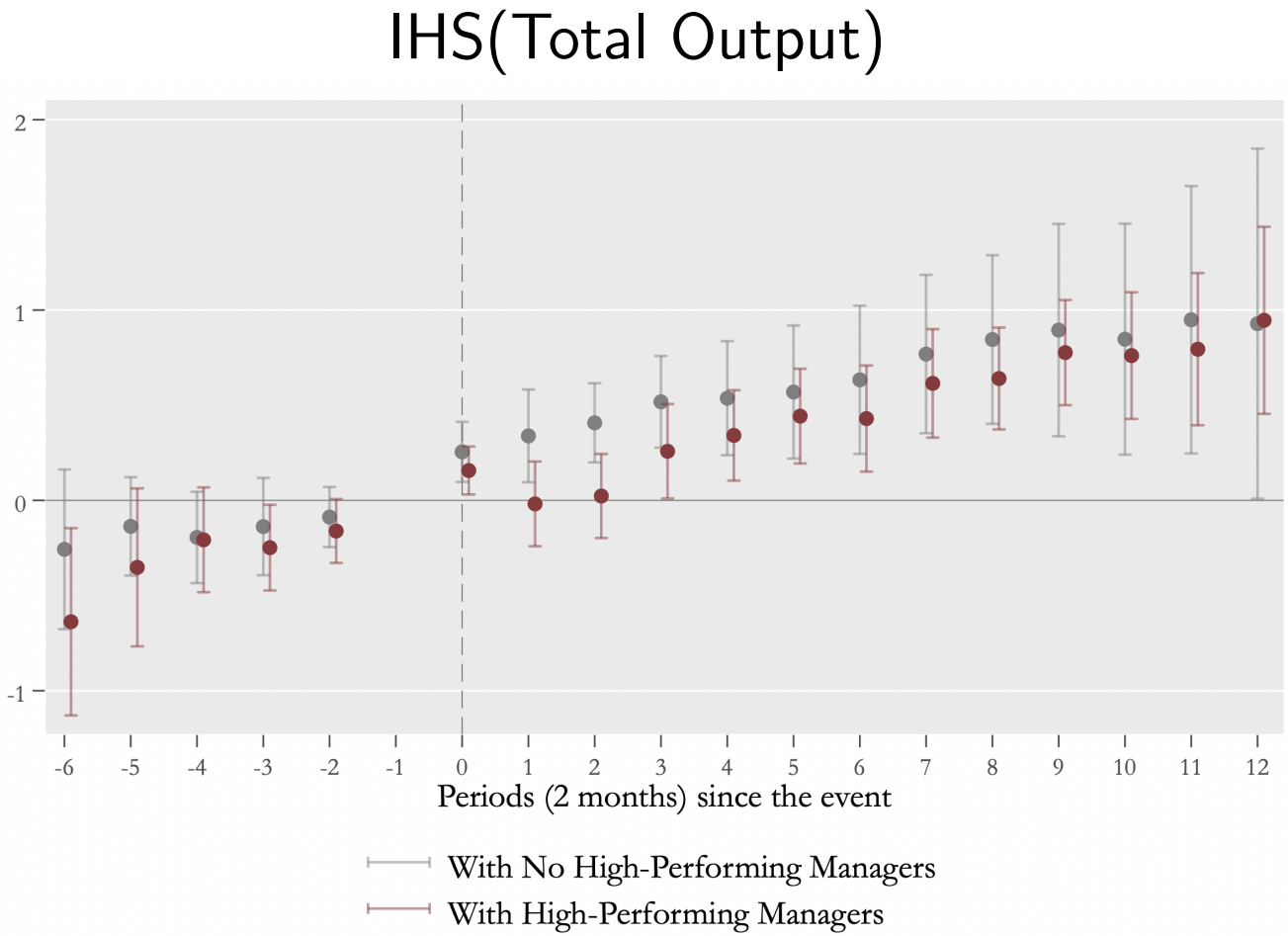
Impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

Back

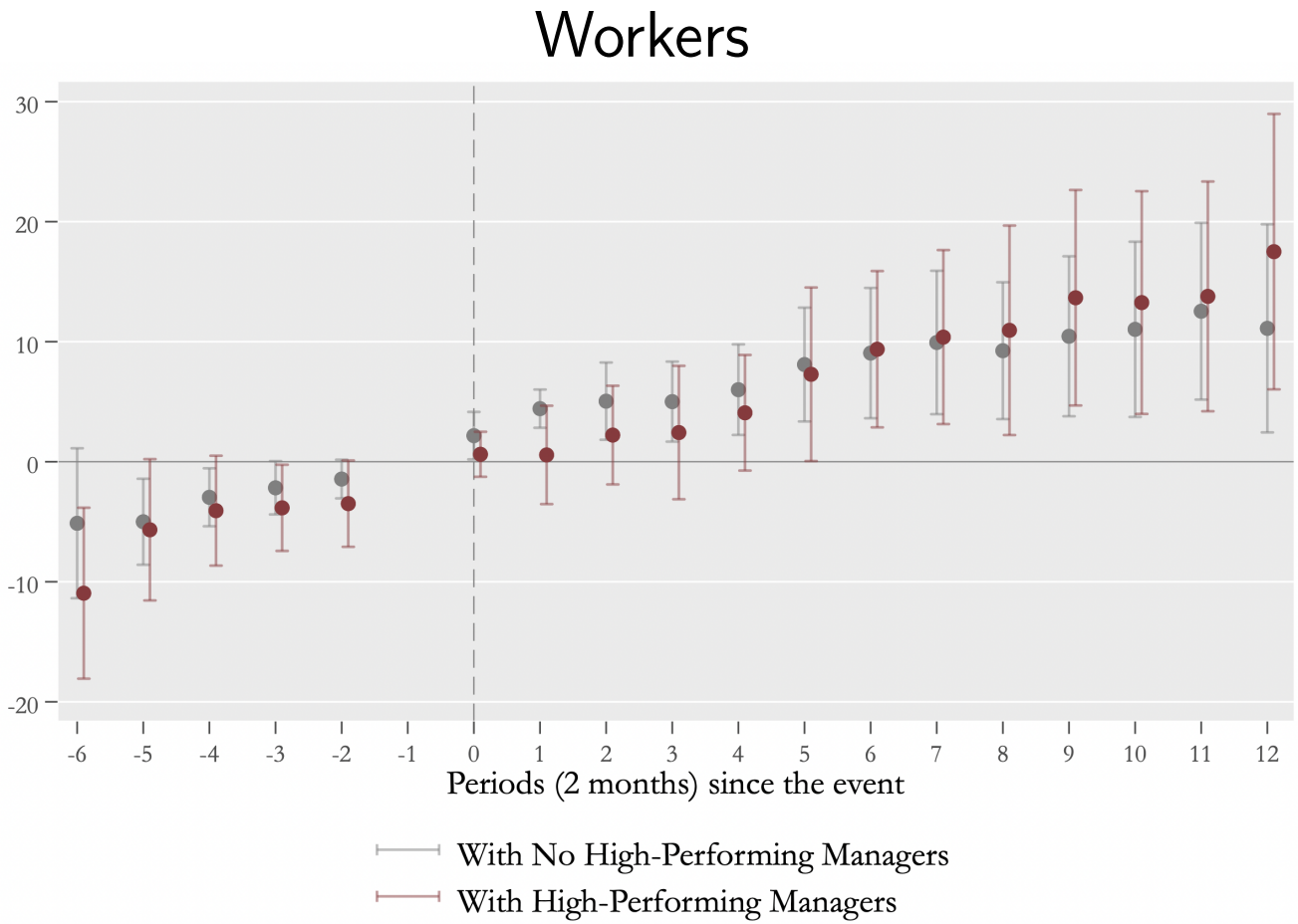
Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

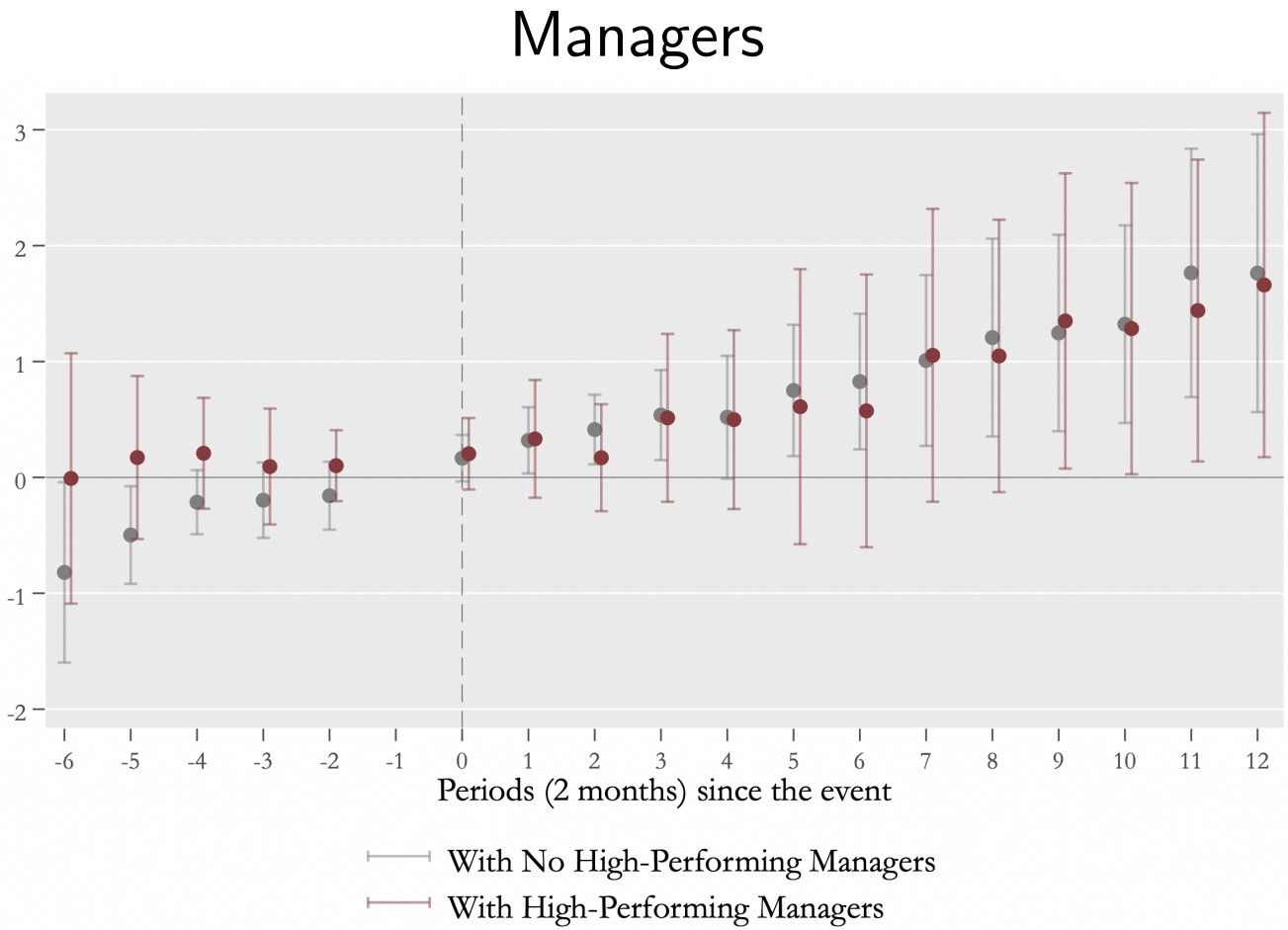
Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

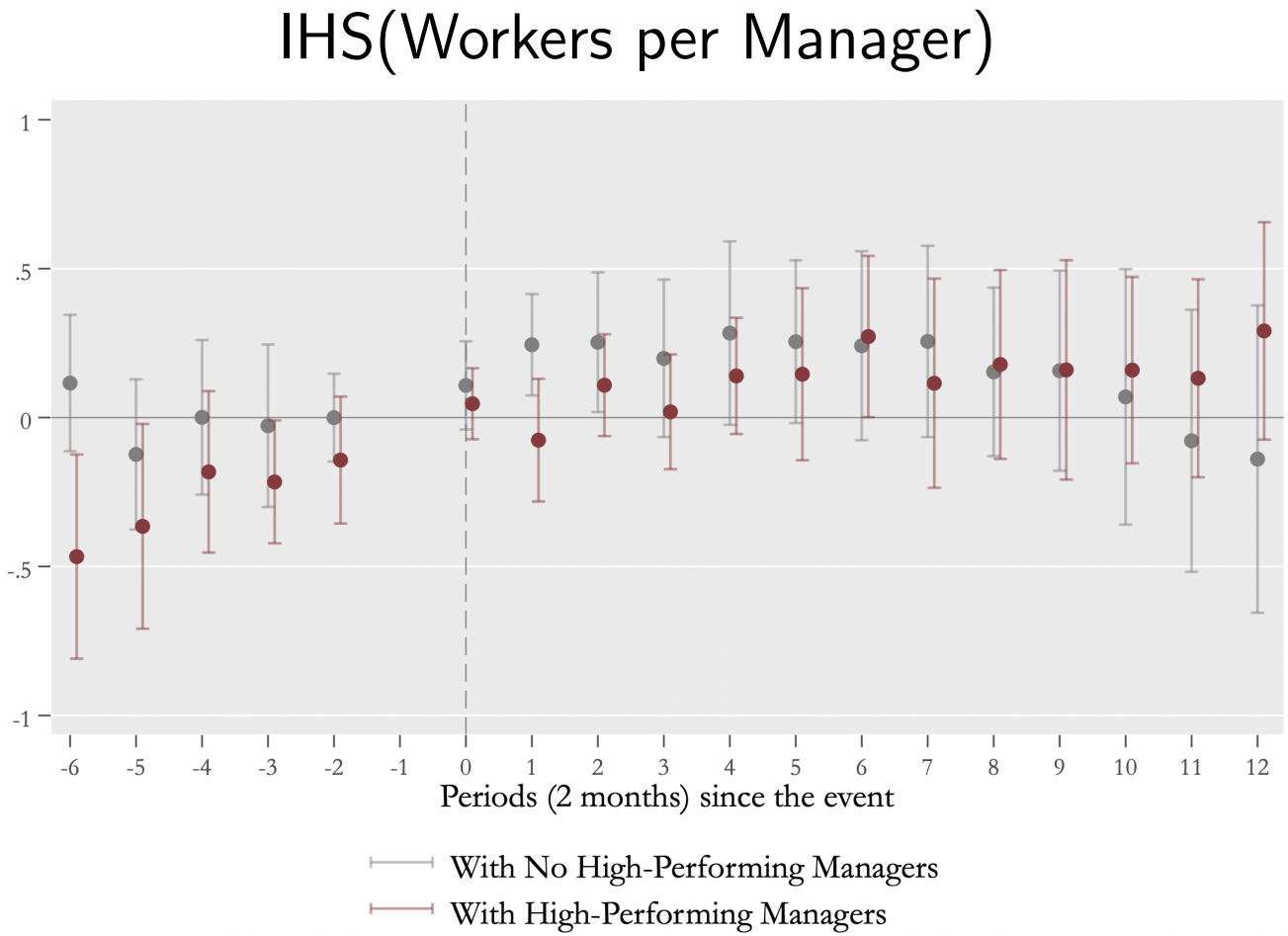
Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

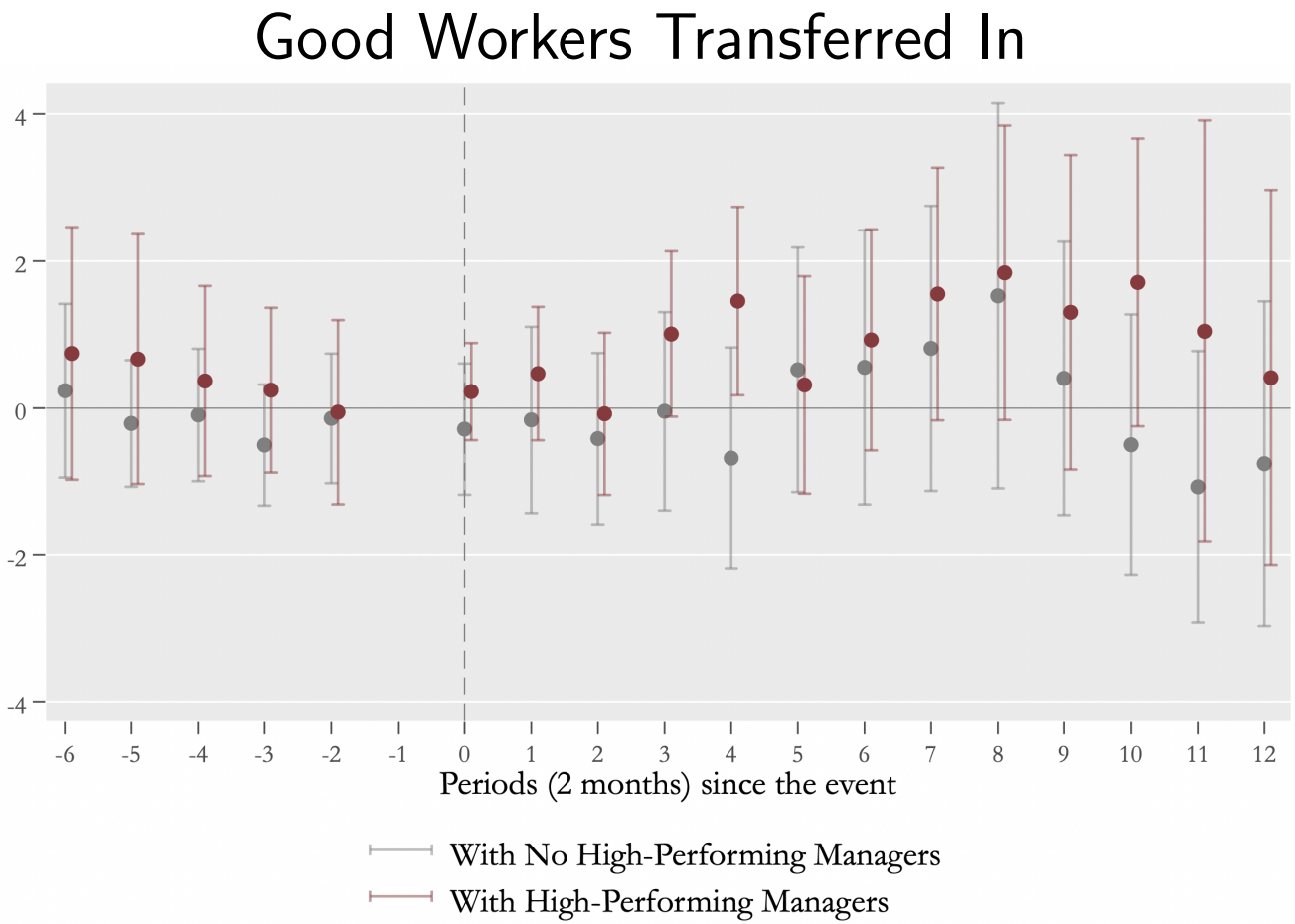
Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

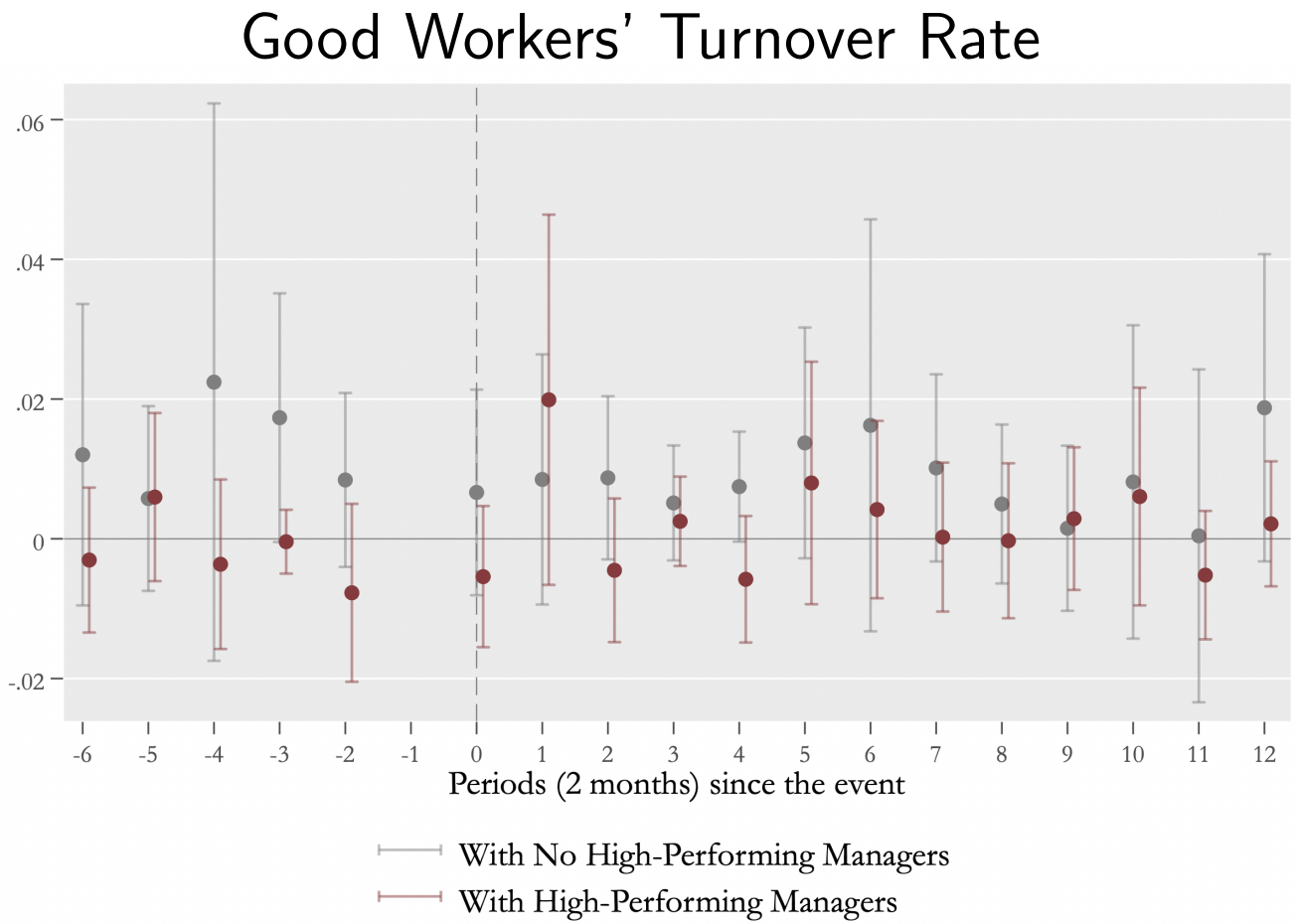
Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

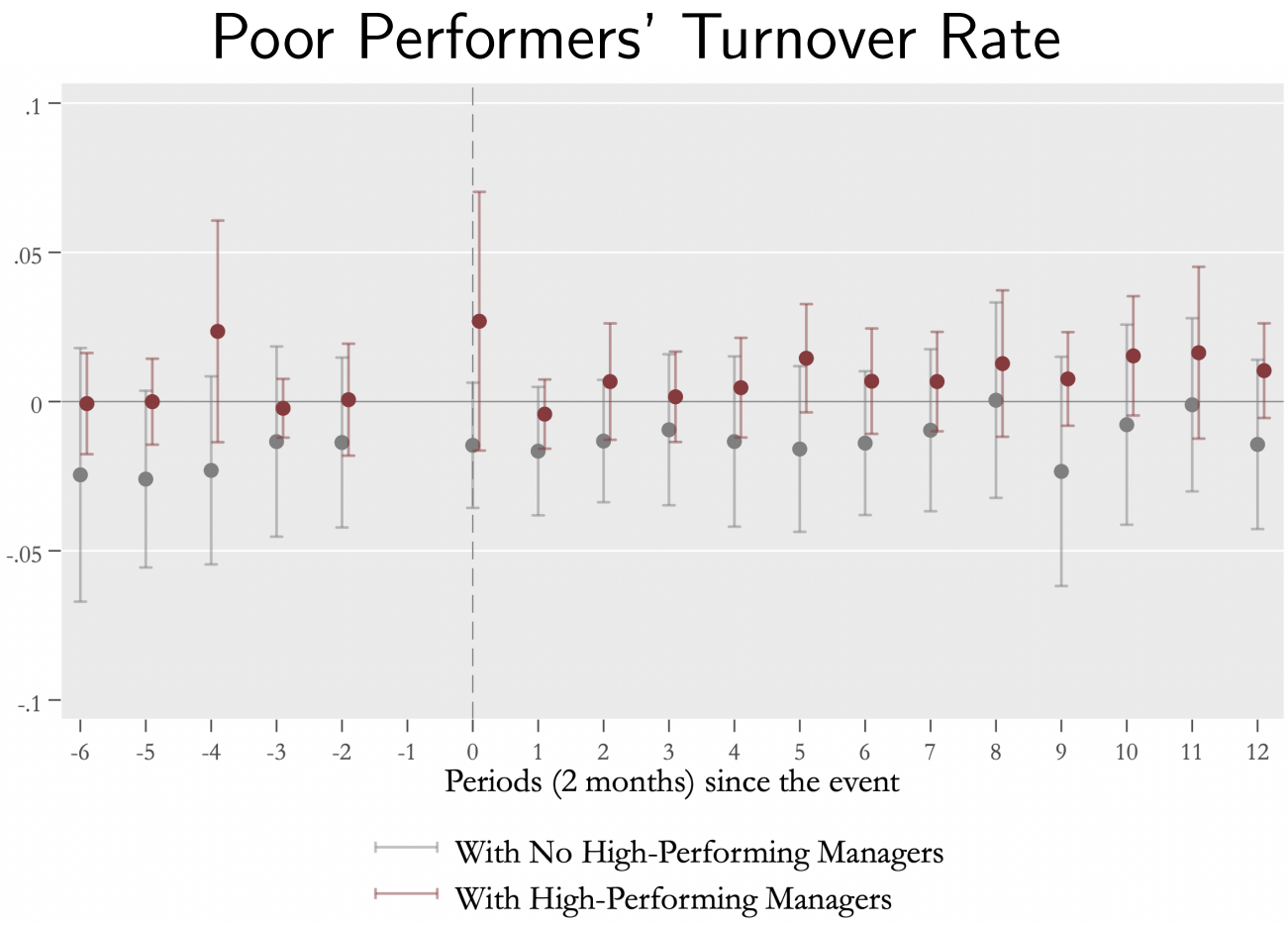
Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

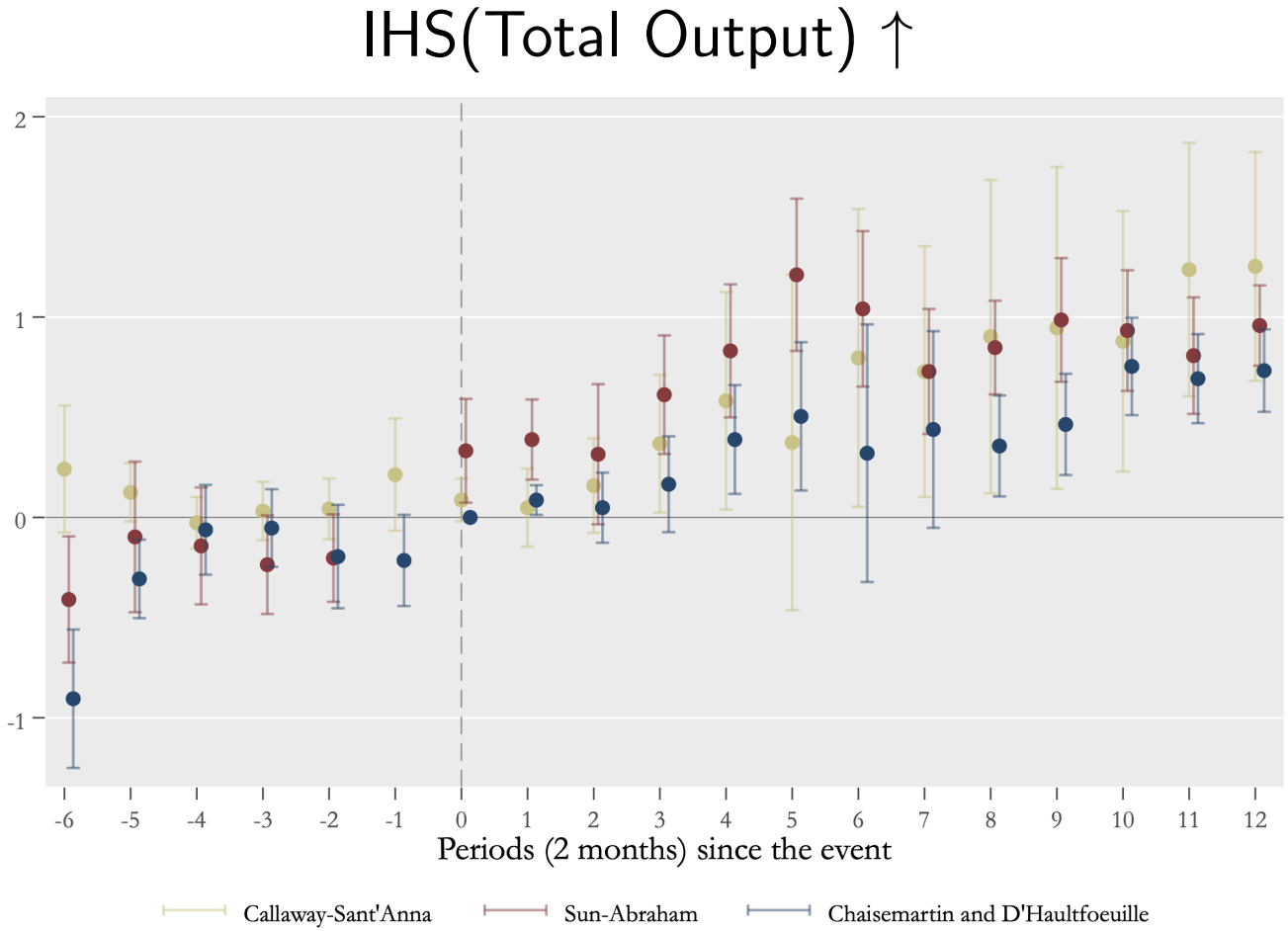
Heterogeneous impacts of implementing the system



Controls for: demographics + team FE + period FE and SE clustered at team level. 955 obs.

▸ Back to Productivity ▸ Back to Good Worker Trans Out

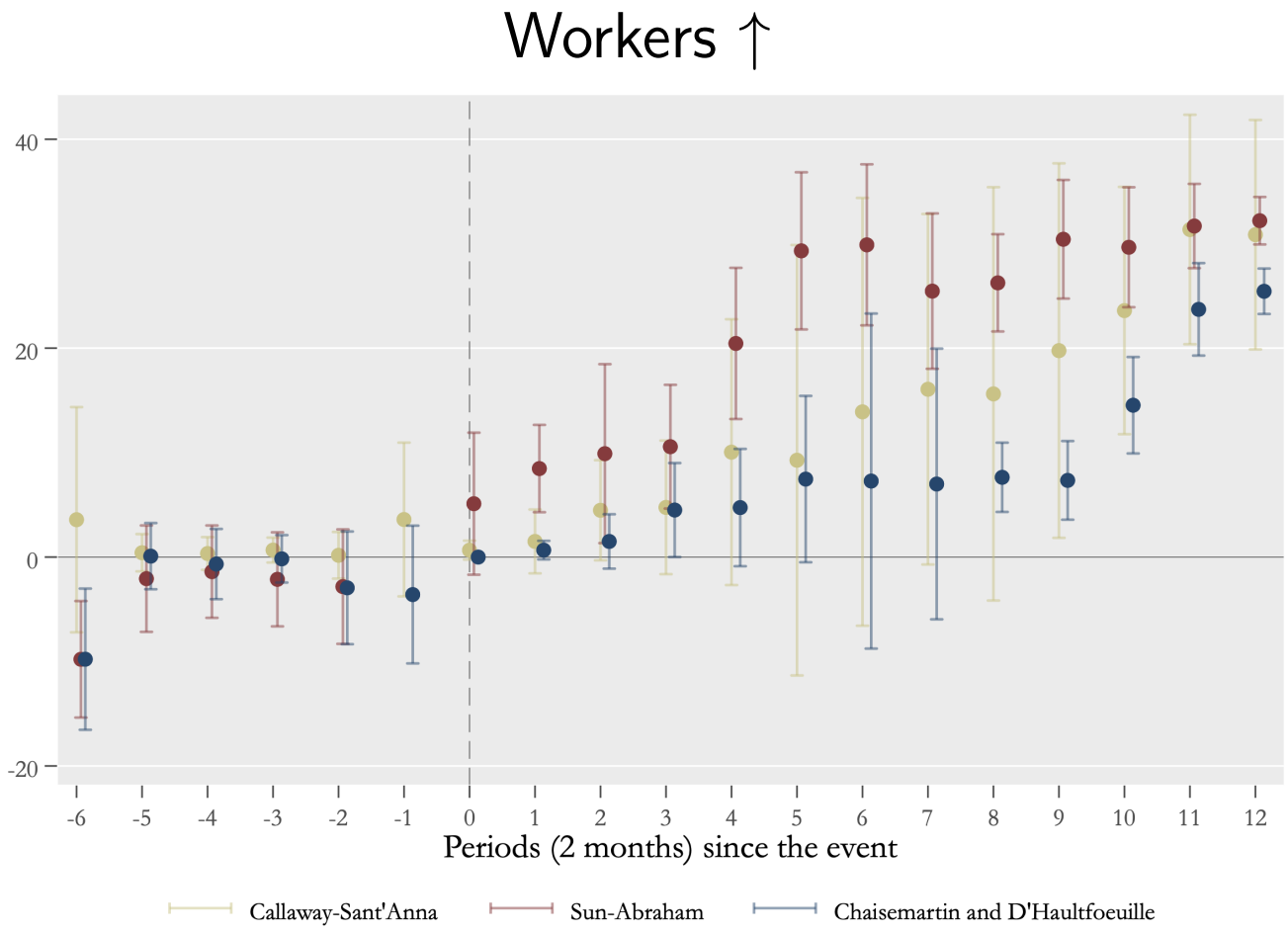
Robustness: DiD corrections



Controls for: demographics + team FE + period FE and SE clustered at team level

Back

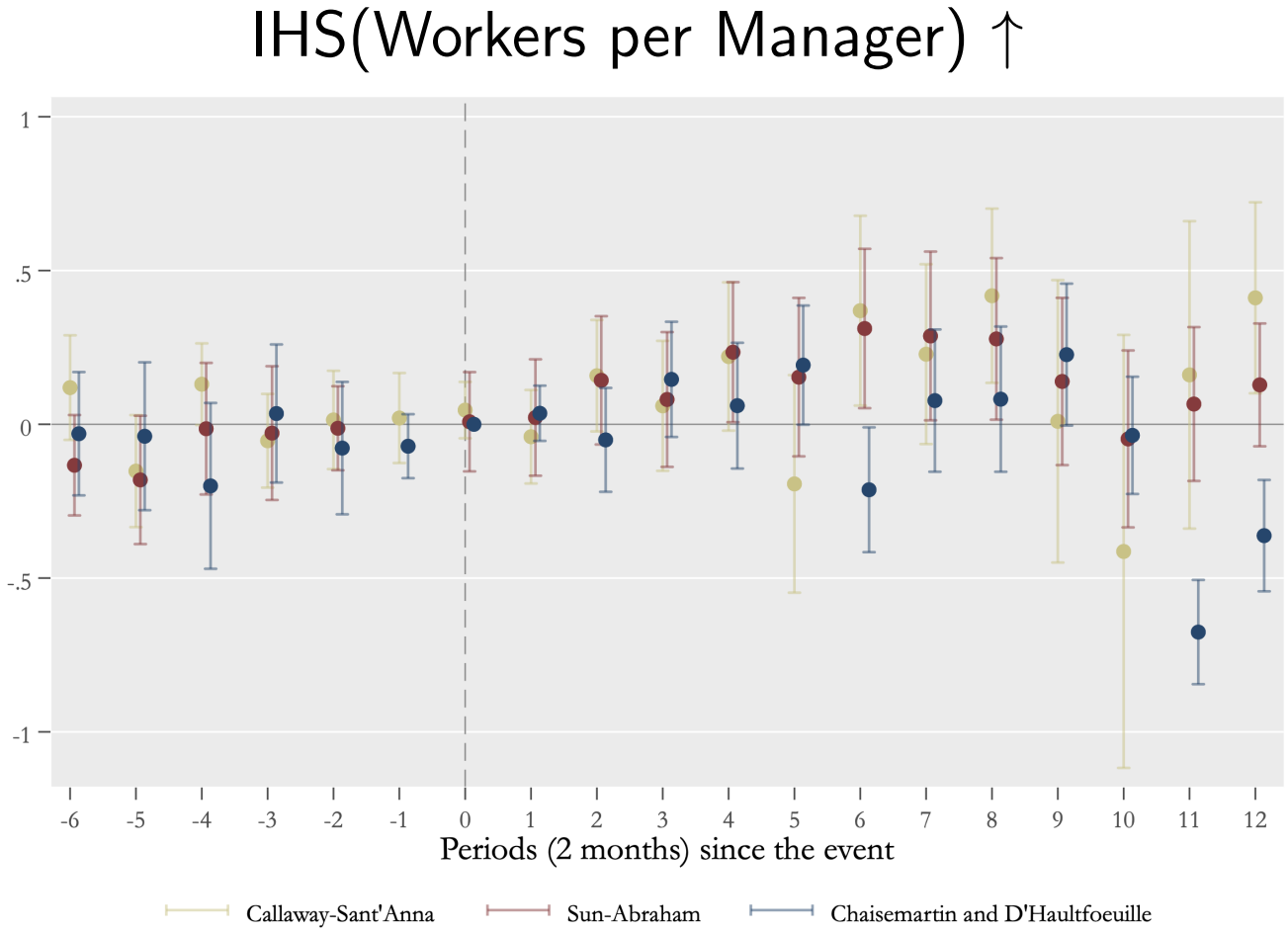
Robustness: DiD corrections



Controls for: demographics + team FE + period FE and SE clustered at team level

Back

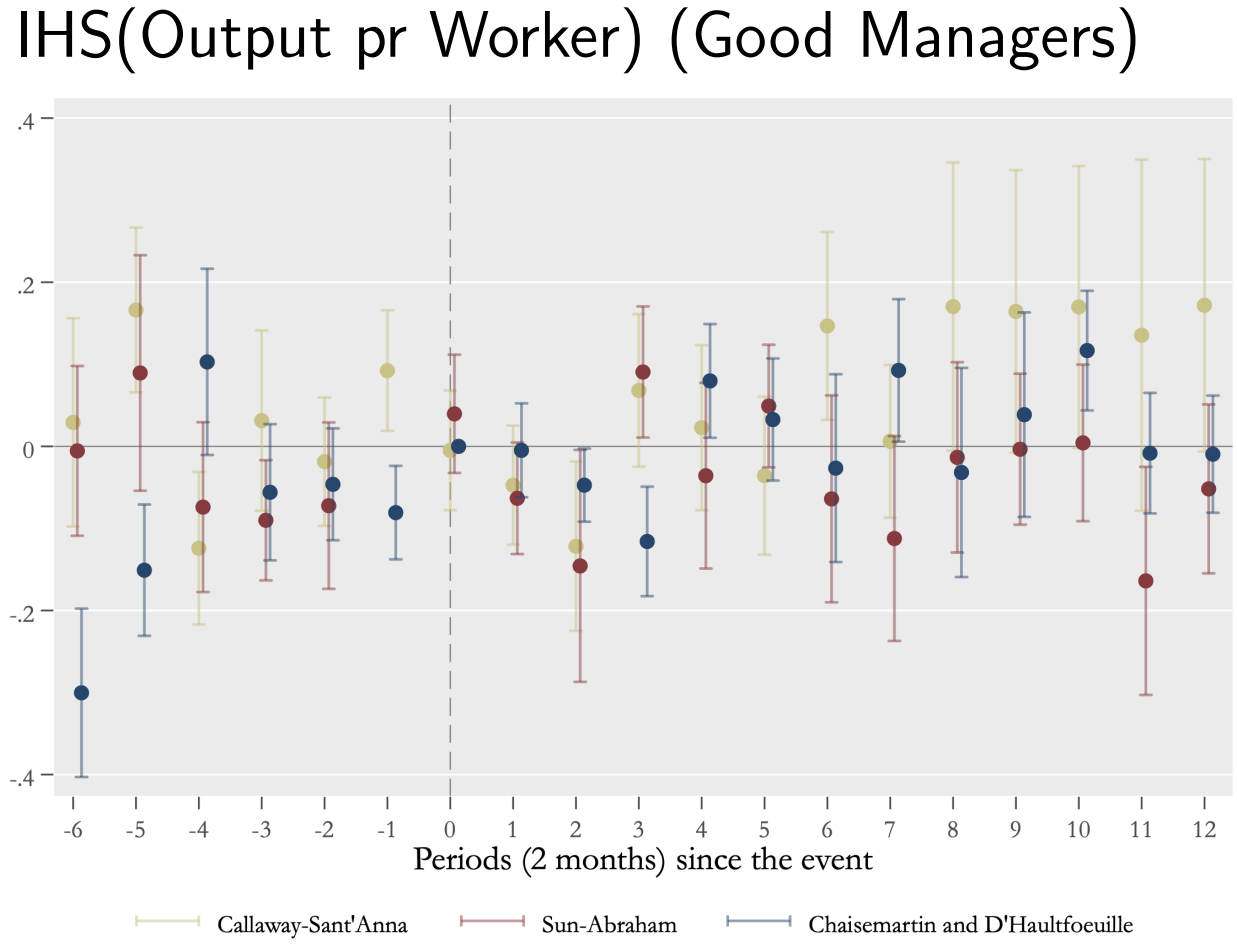
Robustness: DiD corrections



Controls for: demographics + team FE + period FE and SE clustered at team level

Back

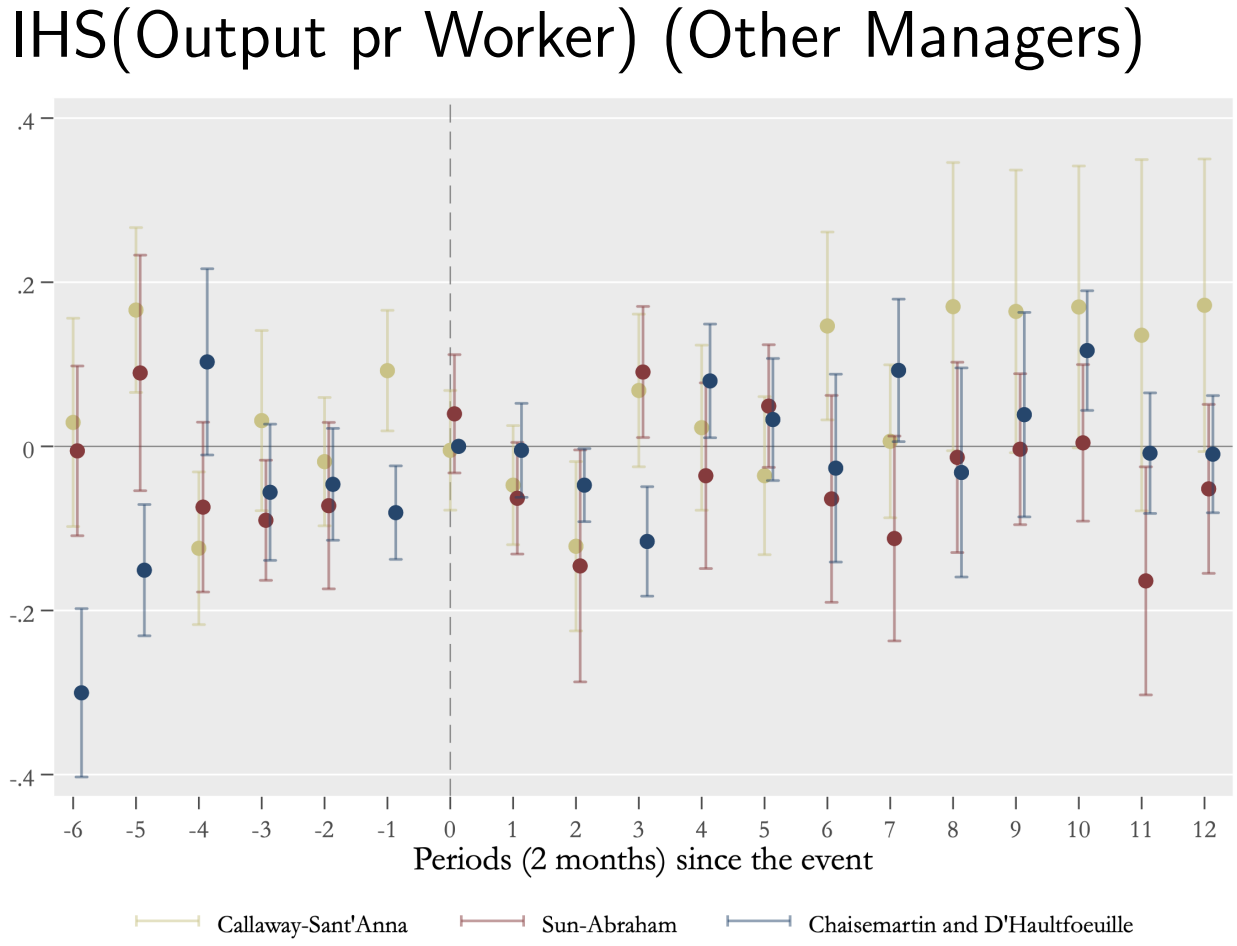
Robustness: DiD corrections for heter. effects



Controls for: demographics + team FE + period FE and SE clustered at team level

Back

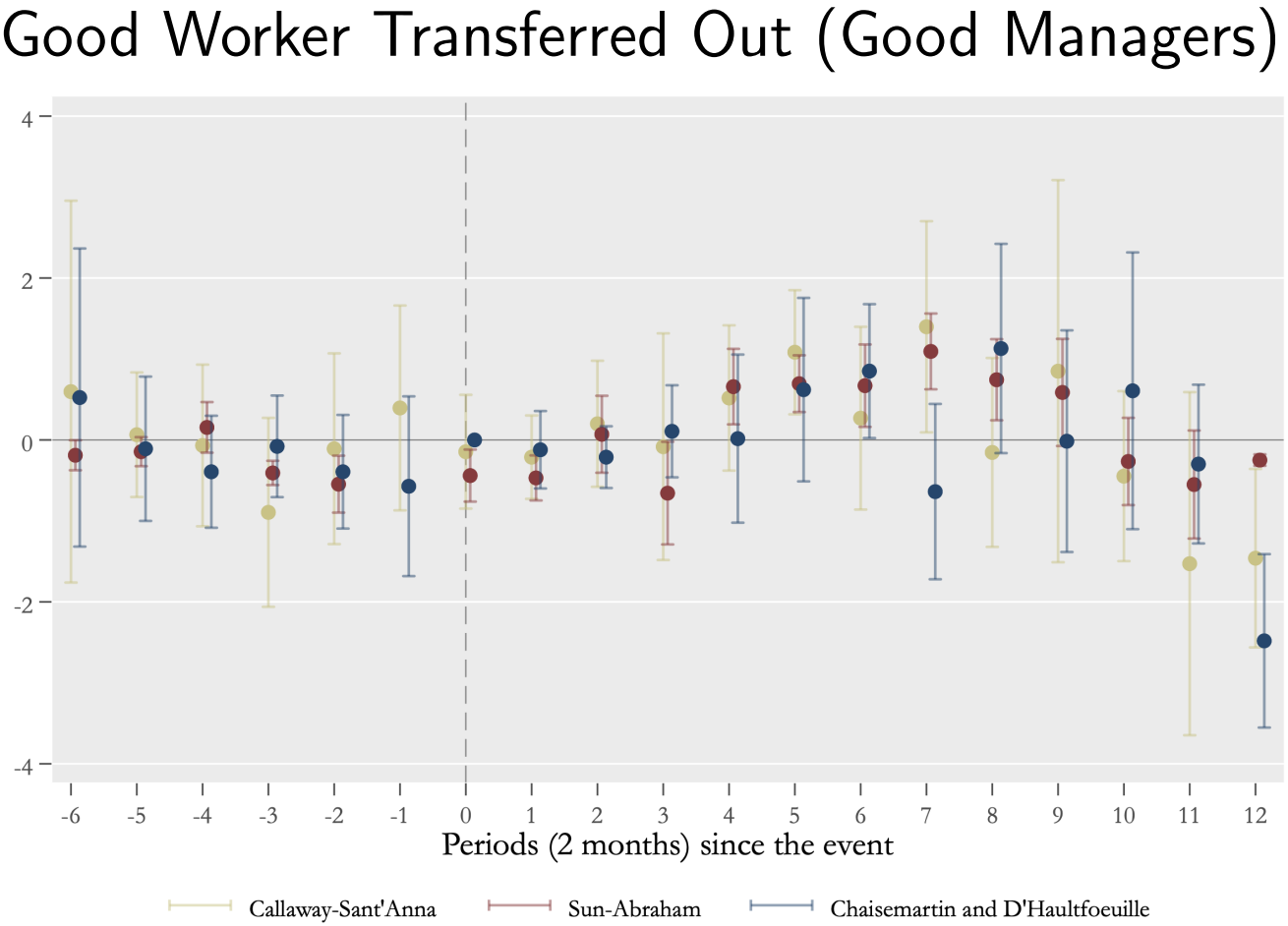
Robustness: DiD corrections for heter. effects



Controls for: demographics + team FE + period FE and SE clustered at team level

Back

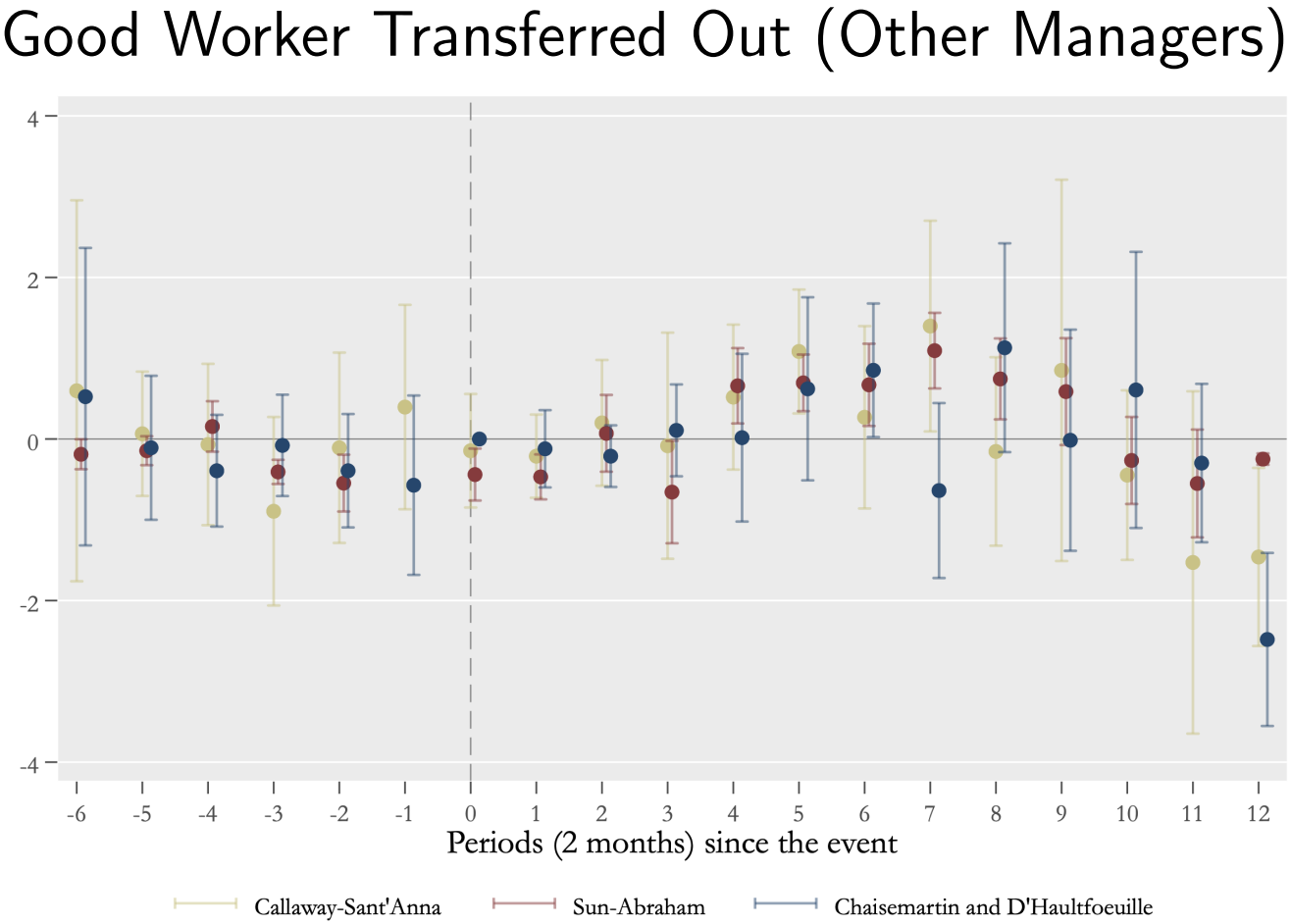
Robustness: DiD corrections for heter. effects



Controls for: demographics + team FE + period FE and SE clustered at team level

Back

Robustness: DiD corrections for heter. effects

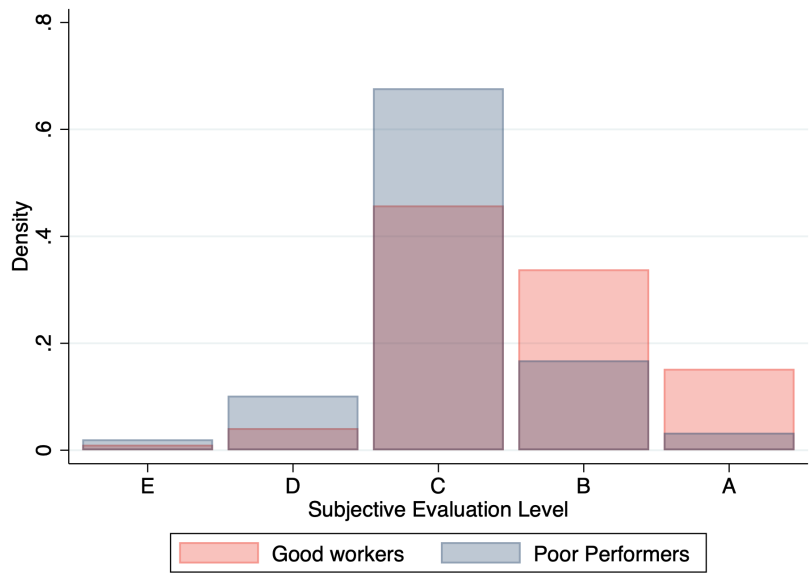


Controls for: demographics + team FE + period FE and SE clustered at team level

Back

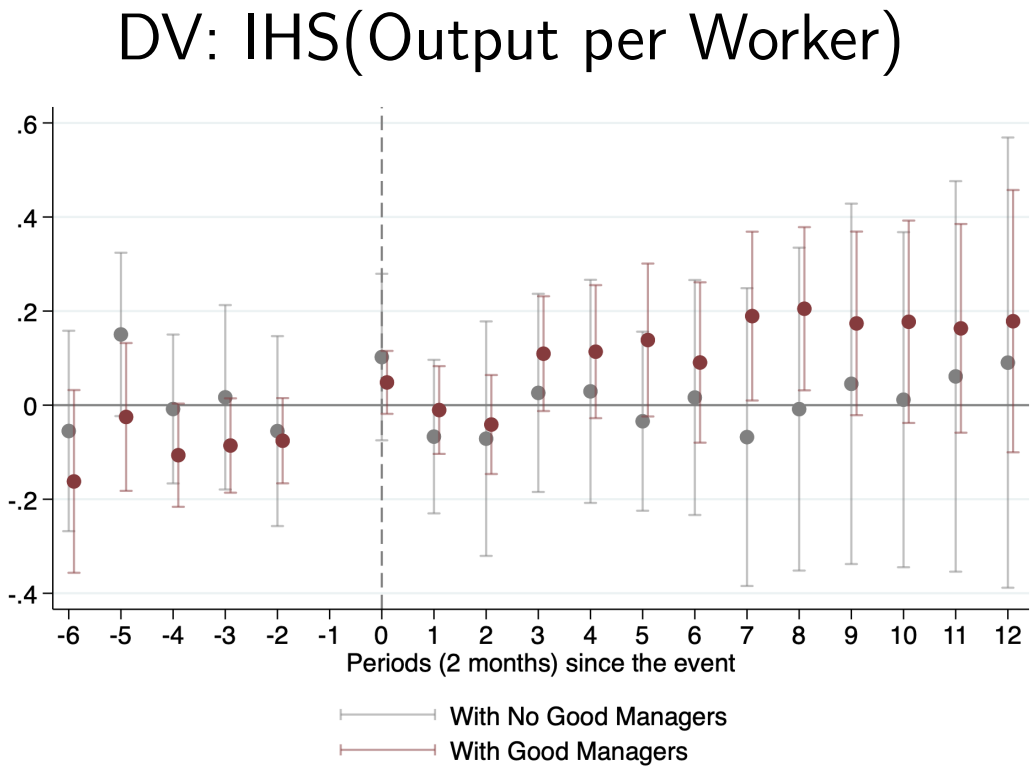
Validation of worker classification

- Good workers are more likely to receive better subjective evaluations than poor performers.



back

Robustness: 50% criteria (event study)



Controls for: demographics + team FE + period FE and SE clustered at team level

Back

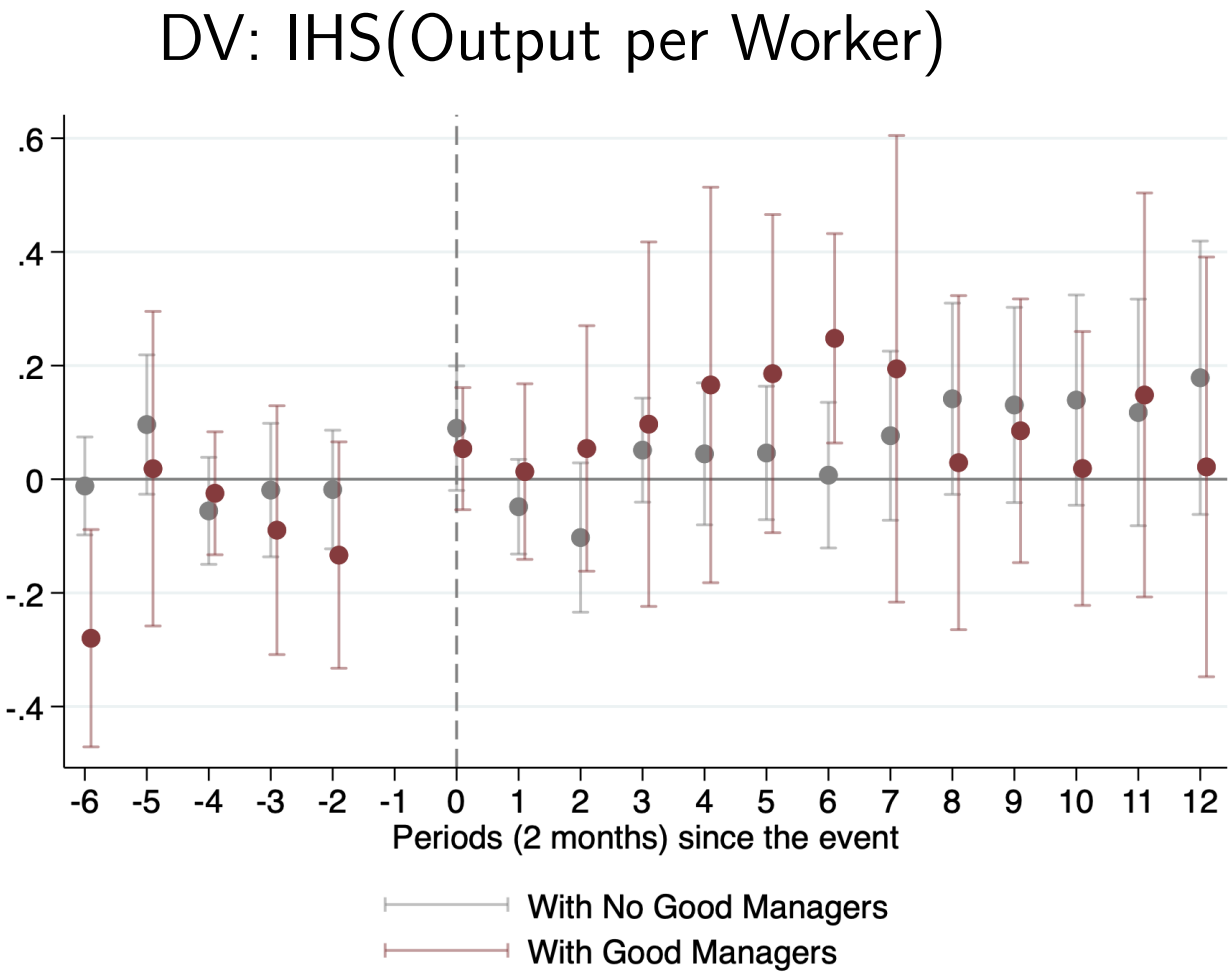
Robustness: 50% criteria (event study)



Controls for: demographics + team FE + period FE and SE clustered at team level

▸ Back

Robustness: manager fixed effects



Controls for: demographics + team FE + period FE and SE clustered at team level

Back

Robustness: manager fixed effects

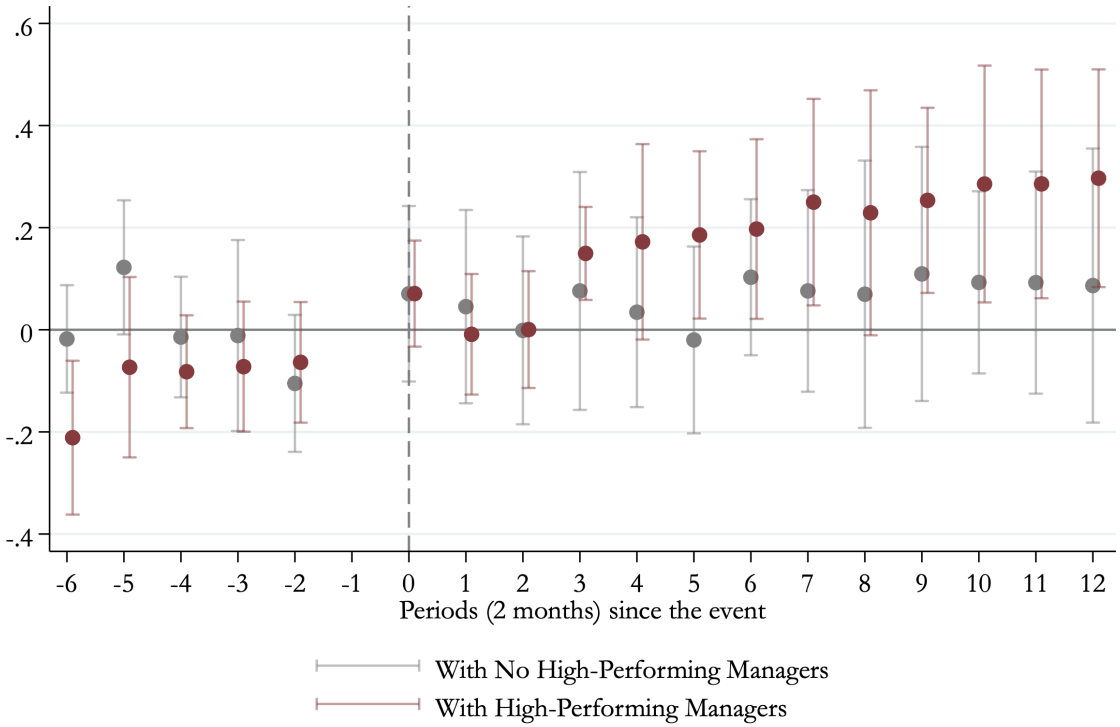


Controls for: demographics + team FE + period FE and SE clustered at team level

Back

Only incumbent workers

Restricting the sample to incumbent workers yields similar result in productivity:



Controls for: demographics + team FE + period FE and SE clustered at team level. 863 obs.

Back

Specification (1)

Our main estimation equation is:

$$Y_{l,t} = \alpha_0 + \sum_{\underline{C} \leq k \leq \overline{C}, k \neq 0} D_{lt}^k \delta_k + \phi_l + \theta_t + X_{l,t} + \epsilon_{l,t} \qquad (1)$$

- $Y_{l,t}$ is the outcome variable
- $D_{l,t}^k$ is a relative time to treatment indicator for whether the team had implemented the system in (bimonthly) period $t - k$
- ϕ_l: team fixed effect
- θ_t: period fixed effect
- $X_{l,t}$: team characteristics
- Standard errors are clustered at team level

▸ back to main impact ▸ back to heter. impact ▸ Back to GW Retention

Specification (2)

Our main estimation equation for ATE is:

$$Y_{l,t} = \alpha_0 + \beta T_{l.t} + \phi_l + \theta_t + X_{l,t} + \epsilon_{l,t} \tag{2}$$

- $Y_{l,t}$ is the outcome variable
- $T_{l,t}$ is an indicator for whether team l has been treated in period t
- ϕ_l: team fixed effect
- θ_t: period fixed effect
- $X_{l,t}$: team characteristics
- Standard errors are clustered at team level

Specification (3)

Our main estimation equation for the heterogeneous ATE is:

$$Y_{k,t} = \alpha_0 + \beta T_{k.t} + \gamma T_{k,t} \times Good_k + X_{kt} + \phi_k + \theta_t^g + \theta_t^b + \epsilon_{k,t} \quad (3)$$

- $Y_{l,t}$ is the outcome variable
- $T_{l,t}$ is an indicator for whether team l has been treated in period t
- $Good_k$ is an indicator for whether team k had at least one good manager before treatment
- ϕ_l: team fixed effect
- θ_t^g: period fixed effects for teams led by good managers
- θ_t^b: period fixed effects for teams led by other managers
- $X_{l,t}$: team characteristics
- Standard errors are clustered at team level

▸ Back to Prod. Boost ▸ Back to GW Retention ▸ Back to 50% Crit. ▸ Back to Cont. Measure ▸ Back to Drop 3t ▸ Back to PW Gap

Specification (4)

Using the worker-day level data, we estimate the following model:

$$Y_{ijt} = \alpha + \beta \text{Good}_{jt} + \phi_i + \xi_t + \epsilon_{ijt} \tag{4}$$

- Y_{ijt} is the dependent variable for worker i managed by manager j on date t
- Good_{jt} indicates if worker i is supervised by a good manager (whether team j has a good manager in period t)
- ϕ_i: individual fixed effects.
- ξ_t: date fixed effects
- Standard errors are clustered at worker level

▸ Back to WL. ▸ Back to real WL. ▸ Back to complexity ▸ Back to adoption & MS

Specification (5)

Using the worker-month level data, we estimate the following model:

$$Y_{ijt} = \alpha_0 + \beta T_{jt} + \gamma T_{jt} \times GM_{jt} + \mu GM_{jt} + \eta_i + \xi_t + \epsilon_{ijt} \qquad (5)$$

- Y_{ijt} is the dependent variable of worker i working in team j in month t
- T_{jt} is a dummy indicating whether team j has implemented the system in month t
- GM_{jt} is a dummy indicating whether team j is managed by a good manager in month t
- η_i: worker fixed effects
- ξ_t: month fixed effects
- Standard errors are clustered at worker level

Back to pay & promote